Classical Mechanics

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Core track, 2011 Fall Modern Physics Stanford Partial
Transcript-derived notes curated by LazyingArt LLC with Video2Book

Original lectures by Leonard Susskind. Transcript-derived course notes curated by LazyingArt LLC with Video2Book.

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Chapter 1

Classical Mechanics and the Meaning of State

Classical mechanics enters this course with an unusually broad claim. Leonard Susskind begins by saying that it is the basis of all physics, not only because it describes ordinary moving objects, but because it provides the general architecture by which physical systems evolve. These companion notes follow that opening lecture closely and preserve a small amount of board evidence, with credit to LazyingArt LLC and Video2Book, when the geometry of the board helps stabilize the mathematical story. The strategy of the lecture is deliberate: before we study realistic particles and forces, we first strip the subject to its simplest logical core and ask what a law of evolution can mean.

1.1 Opening Motivation and the Stroboscopic Toy World

The lecture does not start with Newton's law. It starts with a simplification. Time, for the moment, will not be continuous. We imagine a stroboscopic world in which we look only at equally spaced beats, say one second apart. If q_n denotes the state of the system at the n-th beat, then the law of motion takes the schematic form

$$q_n \mapsto q_{n+1}. \tag{1.1}$$

This is the first important reduction. Instead of worrying about smooth motion, we ask only: given the present state, what is the next one?

The second simplification is to consider systems with only a very small number of configurations. Such systems are less realistic than anything in nature, but that is exactly why they are useful here. They let us separate the structure of the law from the complications of the world to which the law will later be applied.

1.2 Two States, One Law, and the Meaning of Phase Space

The first nontrivial example is a coin on a table. It has two possible configurations, heads and tails. The phase space is therefore

$$\Gamma = \{H, T\}. \tag{1.2}$$

Definition 1.1. The phase space of a system is the space of its possible states.

Definition 1.2. A state is a complete specification of the information needed to determine what happens next.

At this stage the phase space is not ordinary space. It is simply the set of possibilities. The coin gives us two points in an abstract state space, one labelled H and one labelled T.

A useful limiting case helps fix the idea. If there were only one state, then the world would be completely trivial. Nothing could happen, because there would be nowhere else to go. With two states, by contrast, there is room for genuine laws of evolution.

1.2.1 Question & Answer

Question. What exactly counts as a state?

Answer. A state is not just any description we happen to write down. It is exactly enough information to determine the next step. If the symbols H and T are enough, then those symbols are the states. If they are not enough, then the true state space must be larger. This operational definition is the thread that runs through the whole lecture.

1.3 Arrow Diagrams and the First Deterministic Laws

Once the two states are on the board, the lecture introduces its visual grammar. We represent the law by an arrow leading from the present state to the next one. The arrow is not decoration; it is the law of motion itself.

The first law is the dull one:

$$H \mapsto H, \qquad T \mapsto T. \tag{1.3}$$

If the system is in the state H now, it remains in the state H at the next beat. If it is in the state T, it remains in the state T.

The second law is more interesting:

$$H \mapsto T, \qquad T \mapsto H. \tag{1.4}$$

Now the system alternates forever,

$$H \mapsto T \mapsto H \mapsto T \mapsto \cdots . \tag{1.5}$$

These are already deterministic laws in the strong classical sense: if we know the state at one instant, then we know the entire future sequence.

What matters here is not the coin. What matters is that a classical law is being represented as motion in phase space. If we know where we are, we follow the arrow.

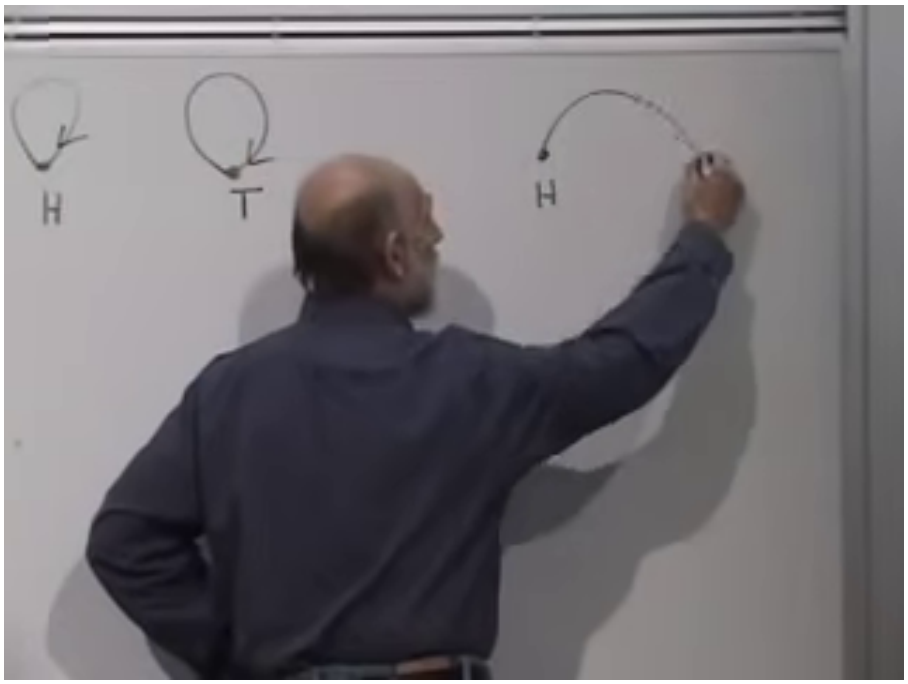


Figure 1.1: Susskind's board as the lecture passes from the trivial self-loop law to the alternating two-state law. The frame directly supports the self-loops and the arrow from H toward T; the reverse arrow is fixed by the transcript.

Figure 1.2: Clean redraw of the two elementary laws on the two-state phase space.

1.4 Allowed and Forbidden Laws in Finite State Spaces

Now the lecture pivots from examples to a criterion. It is no longer enough to draw some arrows. We must ask which arrow diagrams are actually allowed by classical physics.

The first generalization is from two states to six. A die has six faces, so we may imagine six states labelled $1, \ldots, 6$. One possible law is the six-cycle

$$1 \mapsto 2, \qquad 2 \mapsto 3, \qquad 3 \mapsto 4, \qquad 4 \mapsto 5, \qquad 5 \mapsto 6, \qquad 6 \mapsto 1. \tag{1.6}$$

Another acceptable law may break the six states into several disconnected cycles, for example

$$1 \mapsto 1, \qquad 2 \mapsto 3, \qquad 3 \mapsto 4, \qquad 4 \mapsto 2, \qquad 5 \mapsto 6, \qquad 6 \mapsto 5. \tag{1.7}$$

In both cases the rule is deterministic into the future. From each state there is exactly one way to go next.

But classical physics, as the lecture insists, is not only deterministic into the future. It is also deterministic into the past. We must be able to reconstruct where the system came from.

A forbidden example makes the issue plain:

$$1 \mapsto 2, \qquad 2 \mapsto 3, \qquad 3 \mapsto 2.$$

Here the future is unique, but the past is not. If we find ourselves at 2, we cannot tell whether we came from 1 or from 3. Reversing the arrows produces the opposite pathology: then the past is unique but the future is ambiguous.

future unique, past ambiguous past unique, future ambiguous

Figure 1.3: Two forbidden three-state laws. In the first the past is not unique; in the second the future is not unique.

Proposition 1.3. *On a finite phase space Γ, an allowed classical law is a bijection*

$$\sigma : \Gamma \to \Gamma, \tag{1.8}$$

or equivalently a rule with exactly one incoming arrow and exactly one outgoing arrow at each state.

Proof. If a state had two outgoing arrows, the future would not be unique. If it had no outgoing arrow, the system would have no next step. If a state had two incoming arrows, the past would not be unique. If it had no incoming arrow, the motion could not be traced backward through that state. Thus determinism into the future and determinism into the past together require one in-arrow and one out-arrow at every state. On a finite set that is precisely the condition that the update rule be bijective. □

This is the first real mathematical spine of the lecture. A classical finite-state law is a permutation of phase space.

1.4.1 Question & Answer

Question. Why must an allowed classical law have one arrow in and one arrow out at every state?

Answer. Because classical evolution is being characterized here as unique both forward and backward in time. One out-arrow tells us where we go next. One in-arrow tells us where we came from. If either uniqueness fails, information has been lost, and the law no longer has the classical character Susskind is emphasizing.

1.5 Infinite State Spaces, Disconnected Components, and Conservation as Memory

The lecture then removes the assumption that phase space must be finite. Suppose the states are indexed by all integers,

$$\ldots, -2, -1, 0, 1, 2, \ldots,$$

with the law

$$n \mapsto n + 1. \tag{1.9}$$

Each state still has one in-arrow and one out-arrow, but now the trajectory does not cycle through a finite list. It simply marches onward forever.

This already tells us that the finite examples were not important because they were finite. What mattered was the structure of the map on phase space.

The next step is subtler. Phase space may break up into disconnected pieces. One component may be an infinite chain, another a closed cycle, another a fixed point. If the system starts in one component, it never wanders into another. That fact can be rephrased as a conservation law.

This is not yet the conservation of energy or momentum in its familiar form. It is a stripped-down model of the idea of conservation. We may assign a label C to each connected component, for example

$$C = +1 \quad \text{on one component}, \qquad C = -1 \quad \text{on another}, \tag{1.10}$$

and then along the motion we have

$$C_{n+1} = C_n. \tag{1.11}$$

The conserved quantity is simply a memory of where the system started. It never changes because the dynamics never mixes the disconnected parts of phase space.

This is why the lecture identifies reversibility with information conservation. If we know the present state and can reconstruct both the future and the past, then no information about the trajectory has been lost. If two distinct histories collapse into the same present state, information has been destroyed.

At about this point the lecture inserts a valuable warning. Suppose the next coin outcome depended on the previous two entries instead of only the present one. Then the labels H and T would not really be the states after all. The same warning is about to return in continuous form: if position alone does not determine what happens next, then position alone is not the state.

1.6 Continuous Motion and Why Position Alone Is Not Enough

Only after that discrete groundwork does Susskind turn to the real world. He asks for the continuous analogue of the coin question: if a particle moves along a line, is its position enough to determine what happens next?

1.6.1 Question & Answer

Question. Is knowing the position of a particle enough to know what happens next?

Answer. No. We must also know how fast the particle is moving, and in which direction. Position alone is not a complete state. Velocity must be included.

Thus the phase space of a particle on a line is not the line of positions alone. It is the two-dimensional space of pairs

$$(x, v), \qquad v = \dot{x}. \tag{1.12}$$

The horizontal axis is position, the vertical axis is velocity. Positive velocity means motion to the right; negative velocity means motion to the left.

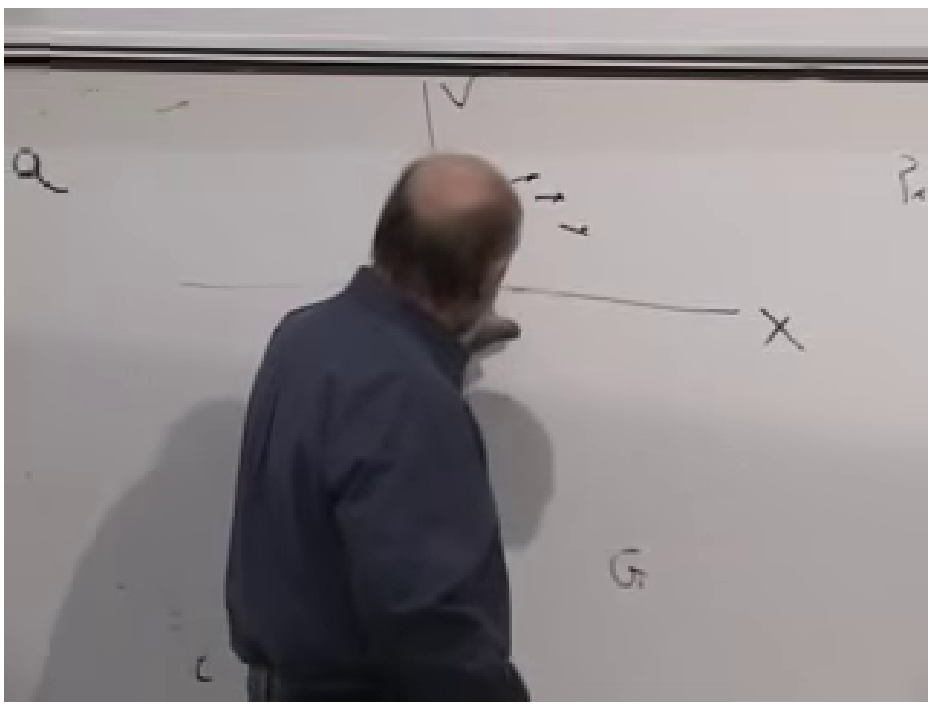


Figure 1.4: Susskind's board near the end of the lecture. The frame supports a local flow picture: a position line, a marked point x, and small arrows indicating the next direction of motion.

The lecture then fills in the meaning of those arrows. If the velocity is zero, the position does not change at the next instant. If the velocity is positive, the next position lies to the right. If the positive velocity is larger, the motion to the right in a fixed time interval is larger. If the velocity is negative, the particle moves to the left.

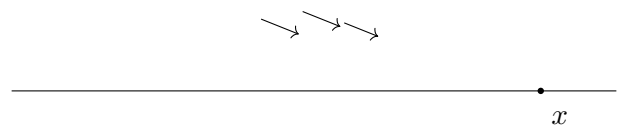


Figure 1.5: A sparse redraw of the local flow sketch. The board frame supports this geometry; the fuller phase-space interpretation comes from the transcript.

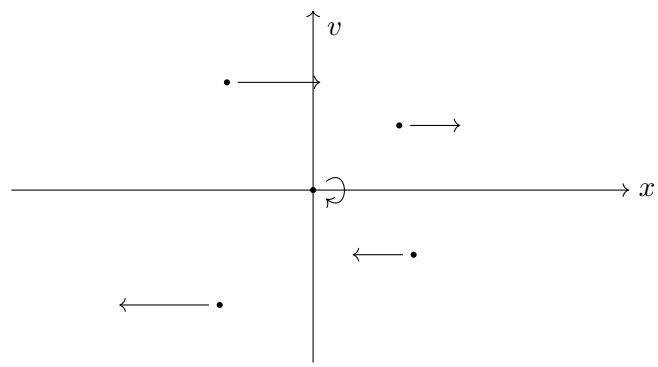


Figure 1.6: Transcript-backed reconstruction of the phase-space picture for motion on a line. Above the x-axis the arrows point right; below it they point left; on the axis itself the next position is unchanged.

The point is no longer merely pictorial. A point in phase space now carries two pieces of information. This is the continuous counterpart of the earlier warning about the coin: whatever information is needed to determine the next step must be built into the definition of state.

The lecture also pauses here over a practical qualification. Classical equations are deterministic in principle, but they are not infinitely predictive in practice. Initial conditions are given by real numbers, and real numbers are never known with infinite precision. Over short intervals the uncertainty may remain small; over longer intervals it may grow, sometimes violently, as in chaotic systems. But the conceptual structure remains the same: for any fixed interval of time, sufficiently precise initial data would, in principle, determine the motion over that interval.

1.7 First-Order Versus Second-Order Equations, and the Return to Two-Step Memory

The next step is to translate the phase-space lesson into the structure of differential equations. The selected board frames do not stabilize these formulas visually, so we write them in the standard notation used throughout the lecture.

A first-order equation contains only first time derivatives. A second-order equation contains accelerations as well:

$$\dot{x} = \frac{dx}{dt}, \qquad \ddot{x} = \frac{d^2x}{dt^2}.$$

To make the logical point as sharply as possible, Susskind first invents a fake law of motion:

$$F(x) = m\dot{x}. \tag{1.13}$$

Assume that the force depends only on the position x. Then if we know x, we know $F(x)$, and the equation immediately tells us $\dot{x}$. Position alone would be enough to determine the next step.

The lecture then differentiates once more:

$$\frac{dF}{dt} = m\ddot{x}, \tag{1.14}$$

$$\frac{dF}{dt} = \frac{dF}{dx}\frac{dx}{dt} = \frac{dF}{dx}\,\dot{x}. \tag{1.15}$$

Since $\dot{x}$ was already known from the first-order equation, $\ddot{x}$ is also determined. Repeating the same idea would determine higher derivatives as well. So a first-order law of this sort would require only one state variable.

But Newton's law does not have that form. Newton's law is

$$F(x) = m\ddot{x}. \tag{1.16}$$

Now if we know the position, we know the force and hence the acceleration. But there is still no equation that tells us the velocity. Therefore position alone does not specify the state. We must supply the velocity as additional information. That is why the phase space is two-dimensional.

This is the same lesson we learned from the coin diagrams, now stated in the language of calculus: the order of the equation measures how much information a state must contain.

1.7.1 Question & Answer

Question. What if the next state depends on the previous two configurations?

Answer. Then a single H or T is not a complete state. The state must be enlarged so that it includes both pieces of information. The law may still be deterministic and reversible, but only on the enlarged phase space.

The lecture returns to the coin example precisely to sharpen this point. If the law depends on the last two entries, then the proper phase space is

$$\Gamma_2 = \{HH, HT, TH, TT\}. \tag{1.17}$$

A clean way to write such a law is to say that the next symbol is a function of the previous pair:

$$(a, b) \mapsto (b, f(a, b)).$$

One reversible completion consistent with the spirit of the classroom discussion is

$$f(H, H) = H, \qquad f(H, T) = T, \qquad f(T, H) = T, \qquad f(T, T) = H,$$

which induces the update rule

$$HH \mapsto HH, \qquad HT \mapsto TT, \qquad TT \mapsto TH, \qquad TH \mapsto HT. \tag{1.18}$$

This is now a deterministic reversible law on four states. The difficulty was never that the law itself was impossible. The difficulty was that the original state space $\{H, T\}$ was too small.

Once that is clear, the higher-order moral follows immediately. If classical mechanics had required positions, velocities, and accelerations as independent initial data, then the equations of motion would have been third order in time and the phase space would have had to include

$$(x, \dot{x}, \ddot{x}).$$

Experiment tells us that ordinary classical mechanics does not work that way. It stops at positions and velocities. For a single particle in three-dimensional space, the phase space therefore has six coordinates: three positions and three velocities.

1.8 Summary

The lecture begins with an intentionally oversized claim and then earns it by working downward to the simplest possible examples. We first learn to think of dynamics as motion in phase space; then we learn that a state is exactly the information needed to know what happens next; then we discover that classical evolution has the special structure of determinism both into the future and into the past.

In the discrete toy world this structure appears as one in-arrow and one out-arrow at every point. In the continuous world it appears as the necessity of using (x, v) rather than x alone, and in the equations of motion it reappears as the difference between a first-order law and Newton's actual second-order law. The lecture's unifying theme is already clear in the first hour: classical mechanics is the study of lawful evolution as conservation of information.

Chapter 2

From Initial Data to Least Action

These notes follow Leonard Susskind's second lecture on classical mechanics, as preserved in the course recording curated by LazyingArt LLC and Video2Book. The lecture opens as review, but it is not idle review. We begin with the question of initial conditions, use a deliberately false law to see what the order of a differential equation means, recover Newton's actual structure by contrast, review energy and momentum in that Newtonian setting, and then turn toward the deeper formulation in which one studies whole trajectories rather than local steps along them.

2.1 First-Order and Second-Order Laws

Susskind begins with one-dimensional motion and a false law of physics. The point is not to rescue Aristotle; it is to understand what a first-order equation would imply if nature had chosen one. The law is

$$F = m\dot{x}, \tag{2.1}$$

or, if the force is known as a function of position,

$$F(x) = m\dot{x}. \tag{2.2}$$

This law would say that in order to keep an object moving one must continuously exert a force on it. Remove the force, and the object comes immediately to rest. That is not our world. But mathematically the law is consistent, and that is enough for the comparison.

Suppose we know the position x at some instant. Then we know the force $F(x)$. But the equation itself then tells us the velocity $\dot{x}$. So for a first-order law of this kind, position already fixes the first time derivative. The natural question is whether it also fixes the higher derivatives. It does.

The force changes with time only because the particle moves through regions where the force differs. Therefore

$$\frac{dF}{dt} = \frac{dF}{dx}\,\dot{x}. \tag{2.3}$$

Differentiating $F(x) = m\dot{x}$ with respect to time gives

$$m\ddot{x} = \frac{dF}{dt} = \frac{dF}{dx}\,\dot{x}. \tag{2.4}$$

So once the position is known, the velocity is known, and once the velocity is known, the acceleration is known. Differentiate once more:

$$m\,\dddot{x} = \frac{d}{dt}\left(\frac{dF}{dx}\,\dot{x}\right) \tag{2.5}$$

$$= \frac{d^2F}{dx^2}\,\dot{x}^{\,2} + \frac{dF}{dx}\,\ddot{x}. \tag{2.6}$$

The jerk is now determined as well. The pattern is obvious enough that Susskind is content to stop there: one can continue indefinitely, and each new time derivative is fixed by the previous ones and by derivatives of the force law. A first-order theory closes from position alone.

Now rerun the same question for Newton:

$$F(x) = m\ddot{x}. \tag{2.7}$$

If we know the position, then we know the force, and therefore we know the acceleration. But the equation says nothing whatever about the velocity. The obstruction is immediate and decisive. A second-order equation does not propagate from position alone. To get started we must specify both

$$x(t_0), \qquad \dot{x}(t_0). \tag{2.8}$$

Only then do we know enough to determine the acceleration, and from there the higher derivatives.

This is the first real lesson of the lecture: Newtonian mechanics is second order in time, and so its initial data are positions and velocities. For many particles the statement is the same in larger dimension: every coordinate and every corresponding velocity component must be supplied.

2.1.1 Question & Answer

Question. Why does Newton need both position and velocity while Aristotle's law did not?

Answer. Because the issue is the order of the differential equation. In the first-order law $F(x) = m\dot{x}$, knowing the position immediately gives the first derivative. In Newton's law $F(x) = m\ddot{x}$, knowing the position gives only the second derivative. The first derivative remains independent, so velocity must be provided as part of the initial data.

2.2 Conservative Forces and Energy Conservation

Having reestablished the initial-data problem, Susskind turns to a quick review of energy conservation. He marks the level of the argument very clearly: the proof is elementary, but the principle is deeper than the proof. Energy conservation survives in electrodynamics, relativity, and quantum mechanics. Here we are only checking that Newton's equations reproduce it in the ordinary conservative case.

He first reminds us that not every force law conserves the mechanical sum of kinetic and potential energy. If we model friction crudely, the kinetic energy appears simply to disappear. Likewise, if some force kept pushing a particle around a circle in the same direction, the particle would come back to the same position with greater speed, so the kinetic energy would increase while the position, and hence any ordinary potential energy, returned to the same value. Such examples are a warning: the familiar energy law is not automatic.

The forces that occur in nature, at least in the Newtonian particle examples under discussion, are conservative. For a single particle they take the form

$$F_i = -\frac{\partial U}{\partial x_i}, \qquad \vec{F} = -\nabla U. \tag{2.9}$$

The minus sign is the content: force points in the direction of decreasing potential energy.

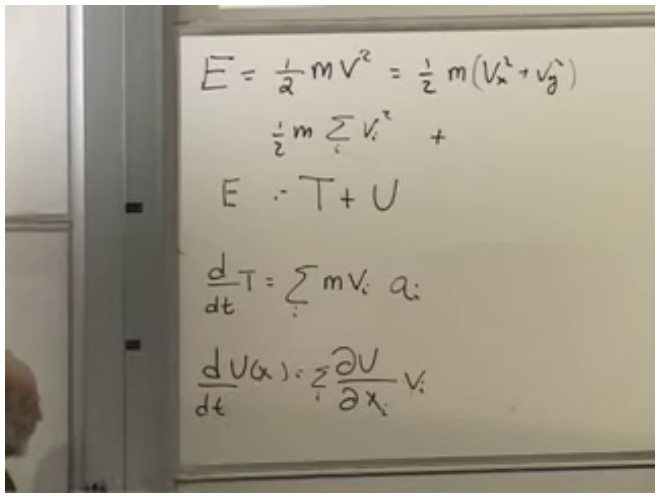


Figure 2.1: Energy definition and separate time-derivative terms. The top handwritten symbol is best read as T, not E; the transcript and the later cancellation make that normalization clear.

Now define the kinetic and total energies:

$$T = \frac{1}{2}mv^2 = \frac{1}{2}m\sum_i v_i^2, \qquad E = T + U. \tag{2.10}$$

The board in the lecture is worth preserving because it shows the order of the argument. First one writes down T, then $E = T + U$, then the separate derivatives of T and U. Only afterward is the cancellation exposed.

The derivative of the kinetic energy is

$$\frac{dT}{dt} = \frac{d}{dt}\left(\frac{1}{2}m\sum_i v_i^2\right) \tag{2.11}$$

$$= \sum_i mv_i\frac{dv_i}{dt} \tag{2.12}$$

$$= \sum_i mv_i a_i. \tag{2.13}$$

The derivative of the potential energy follows from the chain rule:

$$\frac{dU}{dt} = \sum_i \frac{\partial U}{\partial x_i}\frac{dx_i}{dt} \tag{2.14}$$

$$= \sum_i \frac{\partial U}{\partial x_i}v_i. \tag{2.15}$$

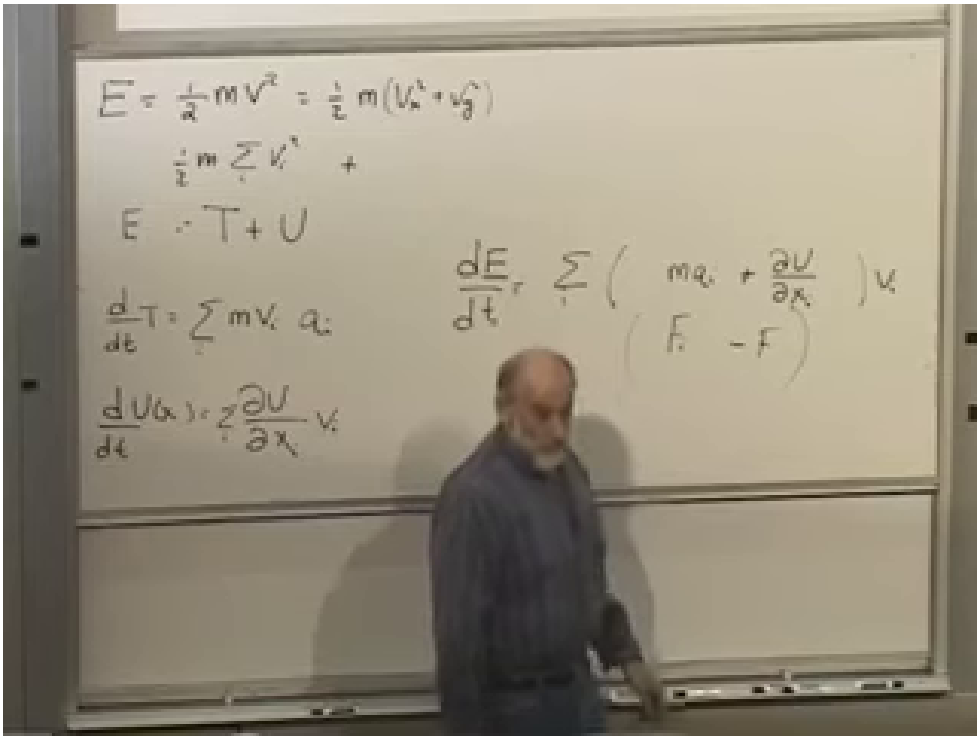


Figure 2.2: Combined time derivative of the total energy and the force-minus-force cancellation.

Now add the two pieces:

$$\frac{dE}{dt} = \frac{dT}{dt} + \frac{dU}{dt} \tag{2.16}$$

$$= \sum_i \left(ma_i + \frac{\partial U}{\partial x_i} \right) v_i. \tag{2.17}$$

At this point the grouping on the board becomes transparent. We write $ma_i = F_i$, while the conservative-force law says

$$\frac{\partial U}{\partial x_i} = -F_i. \tag{2.18}$$

Therefore

$$\frac{dE}{dt} = \sum_i (F_i - F_i) v_i = 0. \tag{2.19}$$

That is the conservation of energy.

The lecture stresses not just the result but the rhythm of the proof. The two derivatives are built separately; only then are they collected into a bracket; and only then do we read the bracket as force minus force. This is review, but it is review with purpose. Already one feels that the conservation law is telling us something more organized than a mere equation of motion.

The same argument extends immediately to many-particle systems. The index i can stand not only for spatial directions but also for different particles. As long as the force on each coordinate is minus the derivative of the potential with respect to that coordinate, the total energy of the whole system is conserved.

2.3 Force as the Time Derivative of Momentum

Susskind now says, in effect, let us come to momentum. Again the mood is review, but the review is widening. Newton's law is rewritten in a slightly different form,

$$F_i = m\frac{dv_i}{dt}. \tag{2.20}$$

Since the mass is constant, we may pull it inside the derivative:

$$F = \frac{d}{dt}(mv). \tag{2.21}$$

And once we give mv a name,

$$\vec{P} = m\vec{v}, \tag{2.22}$$

Newton's law becomes

$$\vec{F} = \frac{d\vec{P}}{dt}. \tag{2.23}$$

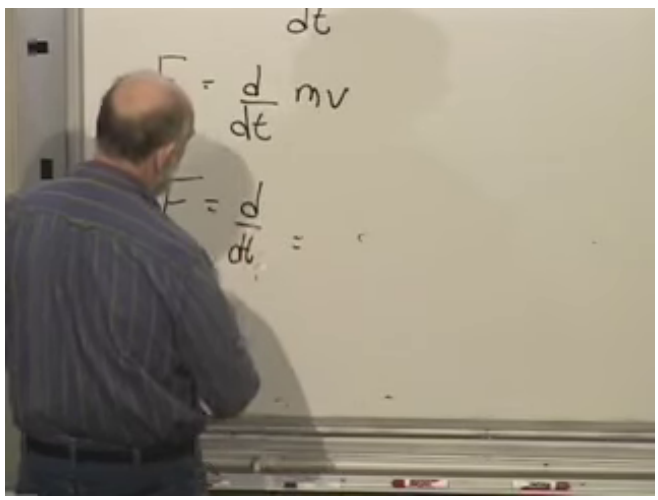


Figure 2.3: Newton's law being rewritten in momentum form. This is a transition frame rather than a finished formula, but it preserves the live progression of the board.

The board hierarchy in the lecture is worth reproducing in clean form:

$$F_i = m\frac{dv_i}{dt}, \tag{2.24}$$

$$F = \frac{d}{dt}(mv), \tag{2.25}$$

$$\vec{F} = \frac{d\vec{P}}{dt}. \tag{2.26}$$

Before going further, the lecture pauses for a sharp question: what assumptions about time are already buried in this Newtonian discussion? The answer is explicit. We are assuming a universal synchronized time, the same at every point in space. We are also assuming, for the moment, that the force law has no explicit time dependence. This is the most naive Newtonian picture, and the lecture wants that simplicity stated out loud before momentum conservation is derived within it.

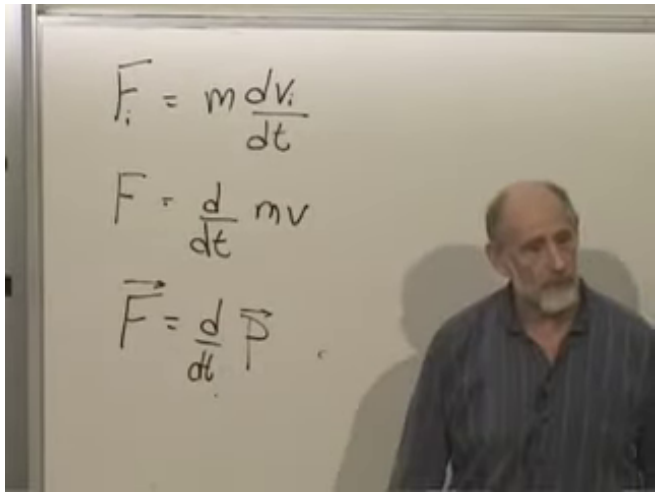


Figure 2.4: Component and vector forms of force and momentum.

Now take a closed system of particles. The force on each particle is the sum of forces due to all the others. For three particles,

$$\vec{F}_1 = \vec{F}_{12} + \vec{F}_{13}, \tag{2.27}$$
$$\vec{F}_2 = \vec{F}_{21} + \vec{F}_{23}, \tag{2.28}$$
$$\vec{F}_3 = \vec{F}_{31} + \vec{F}_{32}. \tag{2.29}$$

The notation is important: $\vec{F}_{12}$ means the force on particle 1 due to particle 2. Newton's third law then says

$$\vec{F}_{12} = -\vec{F}_{21}, \qquad \vec{F}_{13} = -\vec{F}_{31}, \qquad \vec{F}_{32} = -\vec{F}_{23}. \tag{2.30}$$

The last of these corrects a small slip in the transcripted cancellation line; the intended pair is clearly $\vec{F}_{32}$ and $\vec{F}_{23}$.

Since $\dot{\vec{P}}_i = \vec{F}_i$, we have

$$\frac{d\vec{P}_1}{dt} = \vec{F}_{12} + \vec{F}_{13}, \tag{2.31}$$
$$\frac{d\vec{P}_2}{dt} = \vec{F}_{21} + \vec{F}_{23}, \tag{2.32}$$
$$\frac{d\vec{P}_3}{dt} = \vec{F}_{31} + \vec{F}_{32}. \tag{2.33}$$

Add them:

$$\frac{d}{dt}(\vec{P}_1 + \vec{P}_2 + \vec{P}_3) = \vec{F}_{12} + \vec{F}_{13} + \vec{F}_{21} + \vec{F}_{23} + \vec{F}_{31} + \vec{F}_{32} \tag{2.34}$$
$$= 0. \tag{2.35}$$

Hence

$$\frac{d\vec{P}_{\text{tot}}}{dt} = 0. \tag{2.36}$$

So the total momentum of a closed system is conserved.

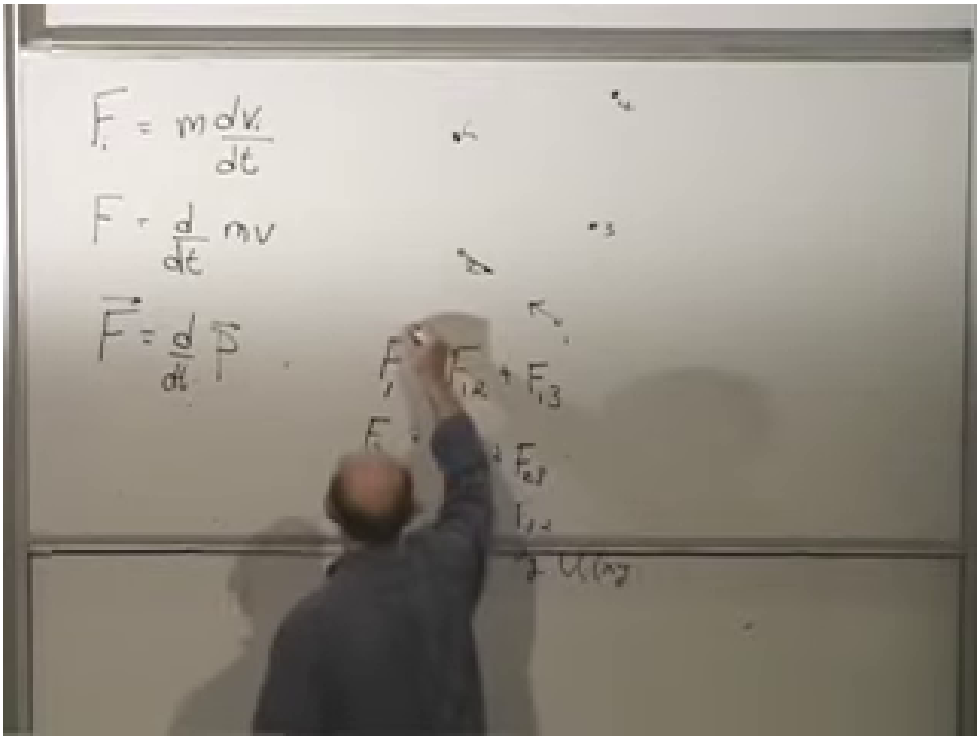


Figure 2.5: Pairwise force sketch for momentum conservation. The exact subscripts in the board drawing are not all legible, so the transcript controls the notation while the image preserves the layout and vector logic.

The lecture immediately widens the scope. Momentum conservation is far more general than this little Newtonian example suggests. It applies to light, to quantum mechanics, to relativity. The derivation we have given uses Newtonian assumptions, but the law itself is deeper.

2.3.1 Question & Answer

Question. Are action and reaction instantaneous, and which part of this argument is specifically Newtonian?

Answer. In Newtonian mechanics, yes: forces are taken to act instantaneously, because time is assumed universal and signal propagation effectively infinite. That is the specifically Newtonian ingredient in the derivation. But the conservation of momentum itself survives when relativity removes instantaneous action at a distance. So the argument is Newtonian; the law is more general.

2.4 A Mathematical Warm-Up: Stationary Points

Only now does the lecture deliberately slow down and say: before we go on to the deep law of classical physics, we need a little mathematics. Not heavy mathematics, but just enough to put everyone back on the same playing field.

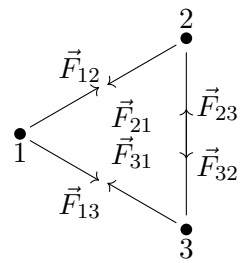


Figure 2.6: A cleaned pairwise-force diagram clarifying the cancellation of internal forces.

Take an ordinary function $f(y)$ of one variable. If we look for a local minimum, the first condition is that the slope vanish:

$$\frac{df}{dy} = 0. \tag{2.37}$$

This condition is not sufficient to distinguish minima from maxima, and it also includes inflection points where the graph is momentarily flat. So the mathematically safer word is not minimum but stationary point. The function does not change to first order under a small change of the variable.

Susskind makes that point explicitly because the principle of least action is really the principle of stationary action. He says he will mostly keep the simpler phrase "least action," but only after flagging the distinction.

For several variables the first-order condition simply splits into one equation per direction:

$$\frac{\partial f}{\partial x_i} = 0 \qquad \text{for every } i. \tag{2.38}$$

If the function is stationary, it must be stationary no matter which direction one moves away from the point.

The lecture then works a concrete two-variable example,

$$f(x, y) = x^2 + y^2 + 3x + 2y + xy + 7. \tag{2.39}$$

The stationary-point equations are

$$\frac{\partial f}{\partial x} = 2x + y + 3 = 0, \tag{2.40}$$

$$\frac{\partial f}{\partial y} = x + 2y + 2 = 0. \tag{2.41}$$

The constant 7 disappears, as it should: an overall vertical shift changes the height of the function but not the location of the stationary point. Solving the equations carefully gives

$$x = -\frac{4}{3}, \qquad y = -\frac{1}{3}. \tag{2.42}$$

The lecture's spoken numbers are somewhat casual, but the equations themselves are stable, and the method is the point.

The transition is now ready. We know how to locate stationary points of ordinary functions. The next step is to study quantities that depend not on finitely many variables but on whole functions, whole curves, whole trajectories.

2.4.1 Question & Answer

Question. Is the principle really least action, or stationary action?

Answer. Strictly speaking, stationary action. What must vanish is the first-order change in the quantity under a small deformation of the path. That condition includes minima, but it can also include maxima or other stationary cases. In the ordinary mechanical examples at hand, one often does have a minimum, and the lecture is content to speak that way, but the more accurate phrase is stationary action.

2.5 Local Motion, Whole Paths, and Functionals

At this point the lecture does something subtle. Before giving geometric examples, it restates the basic problem of mechanics itself. A mechanical system is described by generalized coordinates

$$q_1,\ q_2,\ \ldots,\ q_n, \tag{2.43}$$

which may be Cartesian coordinates, angles, orientations, latitude and longitude, or any other variables that locate the system in configuration space. The full motion is then the collection of functions

$$q_1(t),\ q_2(t),\ \ldots,\ q_n(t). \tag{2.44}$$

That collection is the generalized trajectory.

In the Newtonian formulation, to predict the future we must know the generalized coordinates and their time derivatives at one instant:

$$q_i(t_0), \qquad \dot{q}_i(t_0). \tag{2.45}$$

Newton's equations are local equations along the trajectory. If we know where we are and how we are moving, the equations tell us the acceleration, and so they tell us how to proceed from one point of the trajectory to the next. One does not need knowledge of the whole curve in advance.

The principle of least action will formulate the same mechanics differently. Instead of the position and velocity at one time, we specify the positions at two times, the beginning and the end of the interval. That is a different but equally sufficient way to pose the problem. Susskind emphasizes this with the simple thought experiment of an object whose initial position is known and whose final position after one second is also known. The true trajectory is the one that connects those two endpoints in the allotted time.

Now the geometric examples can do their work. Consider first the shortest curve between two fixed points. Let $y(x)$ describe the curve. For a small segment,

$$ds^2 = dx^2 + dy^2, \tag{2.46}$$

so

$$ds = dx\sqrt{1 + \left(\frac{dy}{dx}\right)^2}. \tag{2.47}$$

Integrating from x_1 to x_2, the total length is

$$s_{12}[y] = \int_{x_1}^{x_2} \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx. \tag{2.48}$$

This is not a function of a few variables. It is a rule that takes an entire function $y(x)$ and returns a number. That is what the lecture now names: a functional.

The lecture insists on both a global and a local viewpoint. Globally, we minimize the total length of the whole curve. Locally, if we focus on a tiny segment between two nearby pinned points, the rule is simply: go straight ahead. The local differential-equation point of view and the global minimization point of view are different ways of expressing the same geometry.

There is also an instructive false shortcut. One might think that since the integrand contains $1 + (dy/dx)^2$, the shortest path should come from setting $dy/dx = 0$. But that would only produce a horizontal line, and a horizontal line will not generally pass through the fixed endpoints. The endpoint condition is part of the problem; one cannot minimize the integrand pointwise while ignoring the global constraint.

The optical example is even closer to what is coming in mechanics. Suppose the speed of light depends on height, $c = c(y)$, as in a layered medium. Then

$$c(y) = \frac{ds}{dt}, \qquad dt = \frac{ds}{c(y)}. \tag{2.49}$$

Substituting the expression for ds,

$$dt = \frac{\sqrt{1 + \left(\frac{dy}{dx}\right)^2}}{c(y)}\, dx. \tag{2.50}$$

Hence the total travel time is

$$t_{12}[y] = \int_{x_1}^{x_2} \frac{\sqrt{1 + \left(\frac{dy}{dx}\right)^2}}{c(y)}\, dx. \tag{2.51}$$

The true light ray is the curve $y(x)$ that makes this time stationary.

The lecture adds one more example in passing: a hanging chain of fixed length chooses the curve that minimizes its potential energy. The exact catenary formula is not derived here, but the point is the same. Physics repeatedly asks us to minimize quantities that depend on whole curves, not just on finitely many coordinates. That is the calculus of variations.

2.6 The Action

Only after this entire runway does Susskind finally say: now let me tell you what the action is. He presents it as something surprising and, even after decades, still surprising. The pedagogical rhythm matters. We are first invited to guess the wrong answer.

Take one particle moving in one dimension, with trajectory $x(t)$. Since the action is to be built from kinetic and potential energy and integrated over time, the naive guess would be

$$\int_{t_0}^{t_1} (T + U)\, dt. \tag{2.52}$$

That is the expected guess. But it is wrong.

The correct quantity is

$$A[x] = \int_{t_0}^{t_1} (T - U)\, dt = \int_{t_0}^{t_1} \left(\frac{1}{2}m\dot{x}^2 - U(x)\right)\, dt. \tag{2.53}$$

This is the action. The integrand is called the Lagrangian:

$$L = T - U. \tag{2.54}$$

At this point the lecture compares the action directly with the earlier functionals. In the least-distance problem the integrand depended on $y(x)$ and on dy/dx. In the least-time problem it depended on $y(x)$, on dy/dx, and on the local light speed $c(y)$. Here, too, the integrand depends on the function and its first derivative: the position $x(t)$ enters through the potential energy $U(x)$, and the derivative $\dot{x}$ enters through the kinetic energy.

This generalizes immediately. For a mechanical system with generalized coordinates $q_i(t)$,

$$L(q, \dot{q}) = T - U, \qquad A[q] = \int_{t_0}^{t_1} L(q, \dot{q})\, dt. \tag{2.55}$$

In Newtonian particle mechanics the Lagrangian is the difference between kinetic and potential energy. In more general systems the Lagrangian may be a different function of q and $\dot{q}$, but the action principle survives. That is why Susskind calls it the deepest law of classical physics.

The lecture also pauses over notation. Since s has already been used for path length, it is awkward to use s for action. The letter does not matter much, but the caution is sensible: the action is not to be confused with the distance functional we have just finished discussing.

The chapter should end where the lecture ends: at the threshold. The action has been revealed, but not yet varied. We have not derived the equations of motion from it. That task is deferred to the next lecture, where the analogy with optics will be completed and the least-action principle will be turned back into differential equations.

2.6.1 Question & Answer

Question. Why is the action built from kinetic minus potential energy rather than from the total energy?

Answer. At the level of this lecture, the honest answer is that $T - U$ is the combination that reproduces Newton's equations when the variational problem is worked out. Susskind explicitly postpones a deeper motivation and prefers first to examine the consequences. The minus sign is not ornamental. Once the potential is defined by $F = -dU/dx$, the sign in the action determines the sign in the resulting equation of motion. The naive integral of $T + U$ does not yield the Newtonian law in the form we want; the integral of $T - U$ does.

2.7 Summary

The lecture begins with a question about initial conditions and ends with the action, but the argument unfolds in a very deliberate sequence. A false first-order law shows what it would mean for position alone to determine the whole motion. Newton's second-order law then shows why velocity must be supplied independently. Energy conservation is reviewed by building its ingredients separately and letting them cancel only at the end. Momentum conservation is derived from pairwise action and reaction, then immediately recognized as deeper than the Newtonian assumptions used in the derivation.

Only after that review does the lecture broaden its scope. We pass from stationary points of ordinary functions to functionals of whole curves, from local differential rules to global variational problems, and from fixed initial data to fixed endpoints. The payoff is the final formula

$$A[q] = \int_{t_0}^{t_1} L(q, \dot{q})\, dt, \qquad L = T - U \tag{2.56}$$

for Newtonian particle mechanics. The next step, and the next lecture's task, is to show how this global principle reproduces the local equations of motion.

Chapter 3

Least Action, Euler–Lagrange, and Symmetry

We begin, as the lecture begins, at a surprisingly large scale. The subject is not merely the motion of one particle, nor even mechanics taken narrowly, but the claim that the laws of physics share a common form. From that wide opening the lecture narrows deliberately: first two small mathematical tools, then the variational derivation itself, and only after that the familiar force laws and conservation laws. The guiding idea throughout is that a global statement about an entire history can be converted into a local equation along the path. The retained board figures come from Leonard Susskind's lecture, with visual curation by LazyingArt LLC and Video2Book, and they are kept only where they witness notation, layout, or geometry that genuinely matters for the mathematics.

3.1 The Common Form of Physical Law

The lecture does not start with $F = ma$. It starts with the broader claim that the laws of physics have a common form. There are mathematical truths, and there are theorems of probability theory, but these are not yet the laws of physics in the specific sense Susskind wants. The laws he has in mind are the dynamical laws: the motion of particles, gravity, electromagnetism, the motion of charged particles, and more generally the machinery by which physical systems evolve.

His claim is that all of these classical laws can be written in a common structural form, and that form is the principle of least action. Even thermodynamics, which is not usually written this way, is placed under the same umbrella: its laws are statistical laws of many degrees of freedom whose underlying microscopic dynamics are themselves governed by an action principle.

This opening matters for the tone of the chapter. Least action is not presented as a decorative reformulation of Newtonian mechanics. It is presented as a common language of physics. Only after making that claim does the lecture step back and say, in effect, before we touch the main argument we need two little theorems.

3.2 Two Mathematical Tools Before the Main Derivation

The first theorem is integration by parts. The transcript is garbled in places here, so we regularize the notation into its standard form while keeping the lecture's actual purpose in view.

We begin with the elementary fact that the definite integral of a derivative is the difference of the function at the endpoints:

$$\int_{t_1}^{t_2} \dot{F}(t)\,dt = F(t_2) - F(t_1). \tag{3.1}$$

Now let $F(t) = f(t)g(t)$. The product rule gives

$$\frac{d}{dt}\big(f(t)g(t)\big) = \dot{f}(t)\,g(t) + f(t)\,\dot{g}(t). \tag{3.2}$$

Integrating from t_1 to t_2, we obtain

$$\int_{t_1}^{t_2} \big(\dot{f}\,g + f\,\dot{g}\big)\,dt = [fg]_{t_1}^{t_2}, \tag{3.3}$$

and therefore

$$\int_{t_1}^{t_2} \dot{f}\,g\,dt = [fg]_{t_1}^{t_2} - \int_{t_1}^{t_2} f\,\dot{g}\,dt. \tag{3.4}$$

This is the form we shall use later. It lets us move the dot from one factor to the other, but only at the price of a minus sign and a boundary term.

3.2.1 Question & Answer

Why is there a dt in the integral? This is exactly the sort of interruption the lecture wants to preserve. The answer is simple and worth stating plainly: an integral must always be an integral with respect to something. In this discussion the independent variable is time, so one must write dt. The expression

$$\int \dot{f}(t)$$

is incomplete notation; the intended expression is

$$\int \dot{f}(t)\,dt.$$

The point is elementary, but the lecture uses it to slow the argument down and make the notation explicit before the harder variational step.

The special case we need repeatedly is the case in which one factor vanishes at both endpoints. Then the boundary term drops out and the formula becomes

$$\int_{t_1}^{t_2} \dot{f}\,g\,dt = -\int_{t_1}^{t_2} f\,\dot{g}\,dt. \tag{3.5}$$

This is exactly the version that will be used when the variation of the path is fixed to vanish at the initial and final times.

The second tool is even smaller, but in the derivation it is decisive.

Proposition 3.1. *If*

$$\int_{t_1}^{t_2} a(t) f(t)\, dt = 0 \tag{3.6}$$

for arbitrary functions $f(t)$, then $a(t) = 0$ on the interval.

Proof. The lecture's argument is the right one. Choose $f(t)$ to vanish everywhere except in a tiny neighborhood of one point t_*. Then the integral is proportional to the value of $a(t)$ near that point. If the integral must vanish for every such choice of f, then $a(t_*) = 0$. Since the point t_* was arbitrary, $a(t)$ must vanish everywhere. □

Together, these two small pieces of mathematics do one large job. Integration by parts moves the derivative, and the arbitrary-function lemma converts an integral statement into a pointwise one. With those in hand, the lecture says we are ready for the big one.

3.3 Trajectories, Histories, and the Local–Global Tension

A system is described by generalized coordinates

$$q_1(t),\ q_2(t),\ \ldots,\ q_n(t).$$

A trajectory, or a history, is the entire collection of these functions of time. To know the history is to know all the q_i for all times in the interval of interest.

The lecture is careful here about how to picture this. In elementary mathematics the independent variable is usually placed horizontally; in physics we often prefer to imagine time running upward. That pictorial choice is secondary, but the point it serves is not. A history is not a single point. It is a path in time.

There are then two ways to formulate the laws of motion.

The first is local. A local law tells us how to move forward from one small piece of the trajectory to the next. Newton's law $F = ma$ is the example Susskind has in mind. If we know the coordinates and velocities at one instant, the differential equation tells us how the path continues.

The second is global. A global formulation looks at the whole trajectory at once. It assigns to that trajectory a quantity called the action, and the true trajectory is the one for which the action is minimized, or more carefully, stationary, among nearby trajectories with the same endpoints.

This is the tension the lecture wants to resolve: how can a global statement about a whole history be equivalent to a local differential law? The motivating answer is simple. If the true trajectory minimizes the action for the whole interval, it must also minimize the action between neighboring points of the path. Otherwise we could replace that local segment by a better one and lower the total action. So the global principle should imply a local equation. That is exactly what the derivation now does.

3.4 Varying the Path and Deriving the Euler–Lagrange Equation

Let $\hat{q}_i(t)$ denote the true path. We deform it by adding a small multiple of an arbitrary function:

$$q_i(t) = \hat{q}_i(t) + \alpha f_i(t). \tag{3.7}$$

Here α is just a number, a parameter measuring how much we have deformed the path. The functions $f_i(t)$ are otherwise arbitrary, except that they must vanish at the endpoints:

$$f_i(t_1) = f_i(t_2) = 0. \tag{3.8}$$

This keeps the initial and final points fixed while allowing the trajectory to vary in between.

At precisely this point the lecture pauses and asks the right question: what is the action whose minimum we are discussing?

Definition 3.2. For a trajectory $q_i(t)$ between t_1 and t_2, the action is

$$A[q] = \int_{t_1}^{t_2} L(q, \dot{q}, t)\, dt. \tag{3.9}$$

In the mechanical examples of this lecture,

$$L = T - U, \tag{3.10}$$

kinetic energy minus potential energy.

The lecture is also careful not to use $L = T - U$ too early in the derivation. For the moment we let L be any function of the coordinates and velocities. The action is then the sum, all along the path, of these little incremental contributions.

Once the path depends on α, the action becomes a function $A(\alpha)$. The true path is the undeformed one, $\alpha = 0$. Since the true path is the stationary path,

$$\left.\frac{dA}{d\alpha}\right|_{\alpha=0} = 0. \tag{3.11}$$

Now differentiate the varied path with respect to α:

$$\frac{\partial q_i}{\partial \alpha} = f_i(t), \qquad \frac{\partial \dot{q}_i}{\partial \alpha} = \dot{f}_i(t). \tag{3.12}$$

Using the chain rule, we obtain the first variation

$$\frac{dA}{d\alpha} = \int_{t_1}^{t_2} \sum_i \left(\frac{\partial L}{\partial q_i} f_i + \frac{\partial L}{\partial \dot{q}_i} \dot{f}_i \right) dt. \tag{3.13}$$

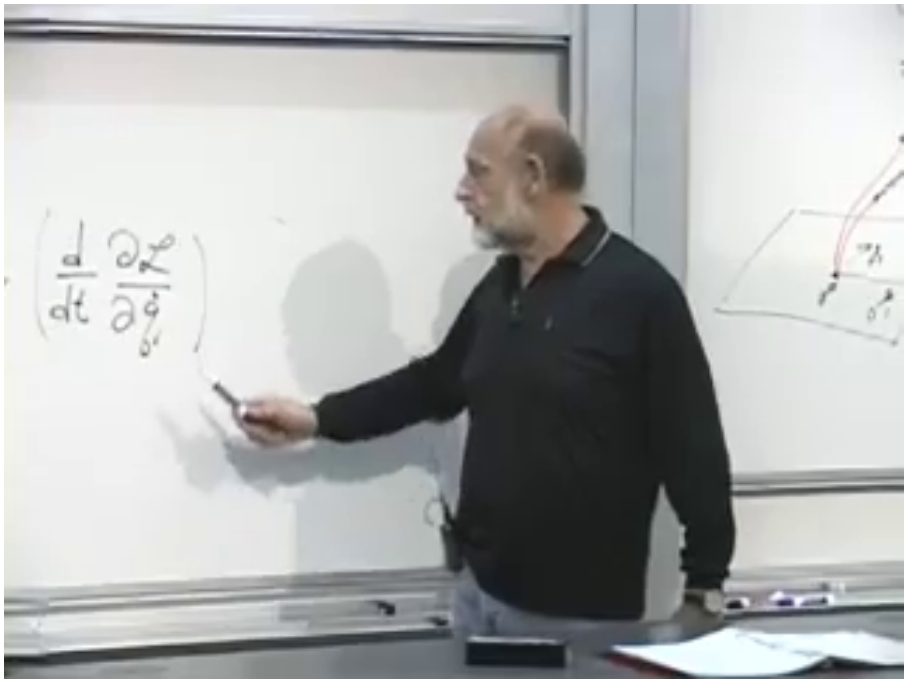
The transcript is noisy at this stage, but the structure is clear: the first term comes from the change in the coordinates, the second from the change in the velocities.

We now use the earlier mathematical tool exactly where it was prepared for use. Integrating the second term by parts,

$$\int_{t_1}^{t_2} \frac{\partial L}{\partial \dot{q}_i} \dot{f}_i\, dt = \left[f_i \frac{\partial L}{\partial \dot{q}_i} \right]_{t_1}^{t_2} - \int_{t_1}^{t_2} f_i \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) dt. \tag{3.14}$$

The boundary term vanishes because every f_i vanishes at t_1 and t_2. Hence

$$\frac{dA}{d\alpha} = \int_{t_1}^{t_2} \sum_i \left[\frac{\partial L}{\partial q_i} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) \right] f_i\, dt. \tag{3.15}$$



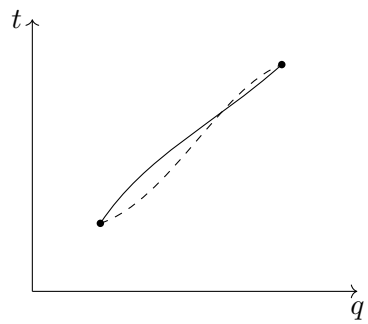


Figure 3.1: The board isolates the time-derivative term in the Euler–Lagrange equation while keeping a nearby trajectory sketch in view. The redraw below is intentionally qualitative: it records only the lecture's local idea of comparing nearby paths between fixed points.

At $\alpha = 0$, stationarity requires this to be zero for every choice of the functions $f_i(t)$. By the arbitrary-function lemma, the bracket itself must vanish pointwise. Therefore

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i}\right) = \frac{\partial L}{\partial q_i}. \tag{3.16}$$

This is the Euler–Lagrange equation.

The lecture's central achievement is exactly this one. A global condition on the whole trajectory has been converted into a local equation that holds at each point along the path.

As it appears on the board, in cleaned notation, the central structure is

$$\frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{q}_i}\right) = \frac{\partial \mathcal{L}}{\partial q_i}. \tag{3.17}$$

In the chapter we use L for the Lagrangian, but the board figure is worth keeping because it witnesses the way Susskind visually separates the algebra from the trajectory picture.

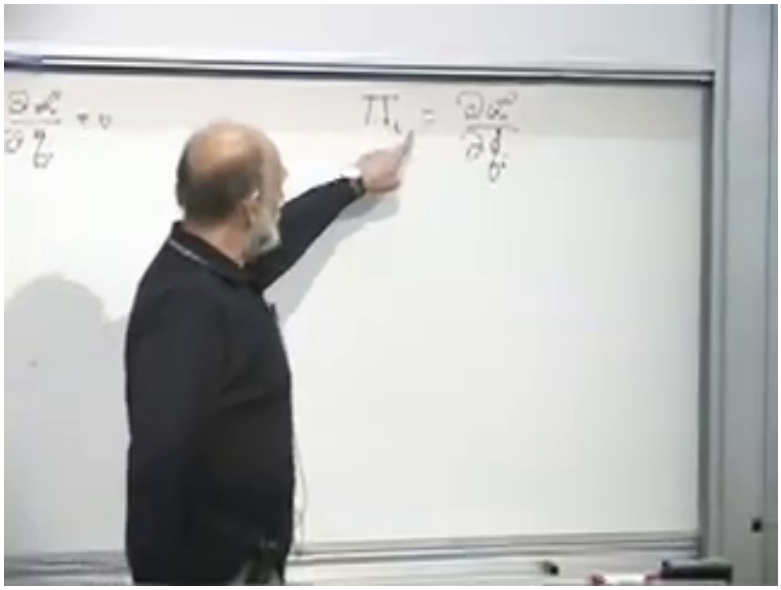


Figure 3.2: The lecture introduces the momentum conjugate to q_i by differentiating the Lagrangian with respect to $\dot{q}_i$. The board symbol can be read as a capital Π_i, but the lecture and standard notation support the lowercase π_i used here.

3.5 What the Euler–Lagrange Equation Says

Once the formal derivation is complete, the lecture immediately turns to language. The new quantities are named only after the structure is in place.

We define

$$\pi_i = \frac{\partial L}{\partial \dot{q}_i}. \tag{3.18}$$

This is the canonical momentum conjugate to q_i. In the board figure the leading symbol is visually a little ambiguous, but both the lecture's language and standard usage support the lowercase π_i, which we keep throughout.

In parallel, the quantity

$$\frac{\partial L}{\partial q_i} \tag{3.19}$$

is called the generalized force. The Euler–Lagrange equation therefore reads

$$\frac{d\pi_i}{dt} = \frac{\partial L}{\partial q_i}. \tag{3.20}$$

Now the equation begins to sound familiar: the time derivative of a momentum equals a force.

3.5.1 Question & Answer

What does the Euler–Lagrange equation say in English? The lecture deliberately postpones this question until after examples, and that postponement is pedagogically right. The first answer is structural: the equation says that a global least-action principle is equivalent to a local differential law along the trajectory.

The second answer is dynamical: once we identify $\partial L/\partial \dot{q}_i$ as momentum and $\partial L/\partial q_i$ as force, the equation becomes a generalized form of $F = ma$. It says that the correct path, singled out globally by the action, is exactly the path whose local motion satisfies a force law.

The quickest way to understand that statement is now to work through examples.

3.6 Worked Newtonian Examples and Translation Symmetry

We start with the simplest possible system: one particle of mass m moving along a line. The Lagrangian is

$$L = \frac{1}{2}m\dot{x}^2 - U(x). \tag{3.21}$$

Here the generalized coordinate is simply $q = x$. Then

$$\frac{\partial L}{\partial \dot{x}} = m\dot{x}, \qquad \frac{\partial L}{\partial x} = -\frac{dU}{dx}. \tag{3.22}$$

Substituting into the Euler–Lagrange equation gives

$$m\ddot{x} = -\frac{dU}{dx}. \tag{3.23}$$

So the variational formalism reproduces the ordinary Newtonian equation of motion.

At once the lecture inserts a useful warning. If we had chosen $L = T + U$, the force law would have come out with the wrong sign. The Lagrangian is $T - U$, not the conserved energy. Energy conservation will be discussed later, and the conserved quantity there will be $T + U$, not L itself.

For many Cartesian coordinates the same calculation simply repeats coordinate by coordinate. If

$$T = \sum_i \frac{1}{2}m_i\dot{x}_i^2, \qquad L = T - U(x_1, \ldots, x_N), \tag{3.24}$$

then

$$p_i = \frac{\partial L}{\partial \dot{x}_i} = m_i\dot{x}_i, \qquad \frac{dp_i}{dt} = -\frac{\partial U}{\partial x_i}. \tag{3.25}$$

The label i now runs over all Cartesian coordinates of all particles. This is still $F = ma$, but now it is written one component at a time for the entire system.

The next example is where the lecture's deeper theme first becomes visible. Consider two particles on a line with a translation-invariant interaction:

$$L = \frac{1}{2}m_1\dot{x}_1^2 + \frac{1}{2}m_2\dot{x}_2^2 - U(x_1 - x_2). \tag{3.26}$$

Translation invariance means that the potential depends only on the separation of the particles, not on where the pair sits on the line as a whole.

To make the force relation explicit, set

$$d = x_1 - x_2.$$

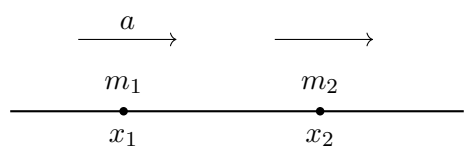


Figure 3.3: A rigid translation moves both particles by the same amount a while leaving the separation $x_1 - x_2$ unchanged.

Then $U = U(d)$, and

$$\frac{\partial U}{\partial x_1} = \frac{dU}{dd}\frac{\partial d}{\partial x_1} = \frac{dU}{dd}, \tag{3.27}$$

$$\frac{\partial U}{\partial x_2} = \frac{dU}{dd}\frac{\partial d}{\partial x_2} = -\frac{dU}{dd}. \tag{3.28}$$

Thus

$$\frac{\partial U}{\partial x_2} = -\frac{\partial U}{\partial x_1}. \tag{3.29}$$

The two Euler–Lagrange equations are

$$\frac{dp_1}{dt} = -\frac{\partial U}{\partial x_1}, \qquad \frac{dp_2}{dt} = -\frac{\partial U}{\partial x_2}. \tag{3.30}$$

Adding them gives

$$\frac{d}{dt}(p_1 + p_2) = 0. \tag{3.31}$$

So the total linear momentum is conserved.

This is the lecture's first fully explicit symmetry-to-conservation argument. Because the action is invariant under a continuous translation, there is a conserved quantity. Here the symmetry is translation invariance, and the conserved quantity is total linear momentum.

The lecture also notes, in passing, that if the potential were made a function of $|x_1 - x_2|$, there would be an inversion symmetry as well. But that is a discrete symmetry, and the present point is the continuous one: symmetries of the action produce conservation laws.

3.7 Two Coordinates, One Broken Symmetry, and Rotational Invariance

The next example is a particle moving near the surface of the Earth. We keep only two coordinates, x horizontally and y vertically. The kinetic term is securely visible on the board, and the transcript gives the potential:

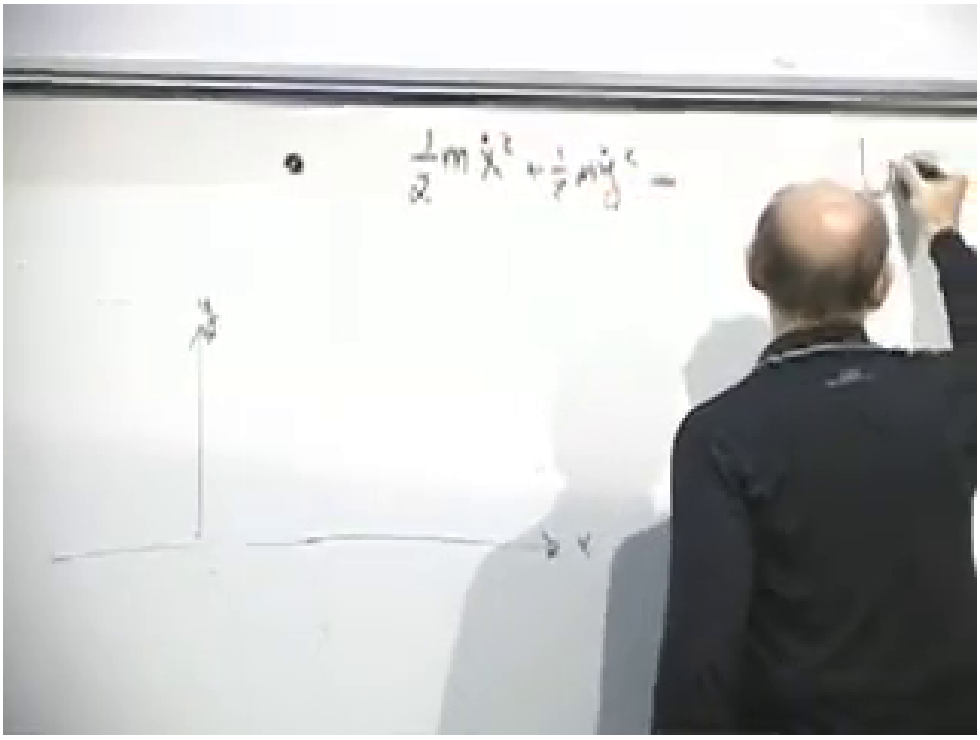
$$T = \frac{1}{2}m\dot{x}^2 + \frac{1}{2}m\dot{y}^2, \qquad U = mgy. \tag{3.32}$$

Thus

$$L = \frac{1}{2}m\dot{x}^2 + \frac{1}{2}m\dot{y}^2 - mgy. \tag{3.33}$$

Because L does not depend explicitly on x, the Euler–Lagrange equation for x gives

$$\frac{d}{dt}(m\dot{x}) = 0. \tag{3.34}$$



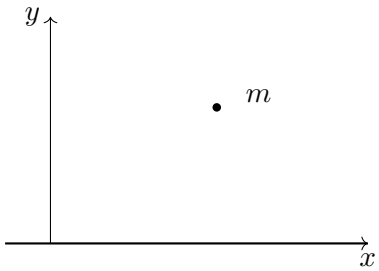


Figure 3.4: The board frame witnesses the two-coordinate kinetic term and the near-Earth geometry. The redraw regularizes only the stable diagrammatic content; the unfinished writing on the far right is intentionally not reconstructed.

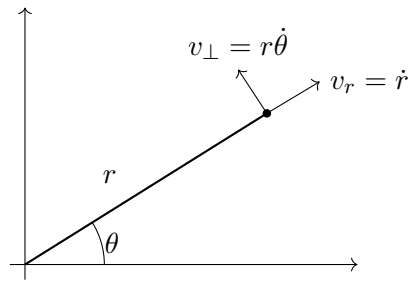


Figure 3.5: Polar-coordinate geometry reconstructed from the lecture's argument. The velocity has a radial component $\dot{r}$ and a perpendicular component $r\dot{\theta}$.

So the horizontal component of momentum is conserved. But for y we have

$$\frac{d}{dt}(m\dot{y}) = -mg. \tag{3.35}$$

There is therefore no conservation of vertical momentum. One symmetry survives; the other does not.

3.7.1 Question & Answer

Why does adding a constant to the potential not restore y-translation symmetry? This is the cleaned mathematical content of the lecture's side discussion. We may always add a constant to U without changing the physics, because the force depends only on derivatives of U. But translating the coordinate y is a different operation. If $U = mgy$, then changing y changes U itself, not merely by a once-and-for-all harmless offset tied to the chosen zero of energy, but by a change that depends on the new value of y. So the Lagrangian is not invariant under y-translations. That is why there is no conserved y-momentum.

This example sets up the lecture's next pivot. The principle of least action is not tied to rectangular coordinates. If we choose another coordinate system, the formulas change clothes, but the action principle itself does not change.

So let us pass to polar coordinates for a particle moving in a plane. The coordinates are r and θ, and the trajectory is now the pair $r(t)$, $\theta(t)$. The lecture derives the kinetic energy geometrically by decomposing the velocity into a radial piece and a perpendicular piece:

$$v_r = \dot{r}, \qquad v_\perp = r\dot{\theta}. \tag{3.36}$$

Therefore

$$T = \frac{1}{2}m\left(\dot{r}^2 + r^2\dot{\theta}^2\right). \tag{3.37}$$

Now take a central potential,

$$U = U(r), \tag{3.38}$$

so that

$$L = \frac{1}{2}m\left(\dot{r}^2 + r^2\dot{\theta}^2\right) - U(r). \tag{3.39}$$

The transcript becomes noisy in the radial equation, and the lecture's stable emphasis lies elsewhere. The important point is that L does not depend explicitly on θ. Therefore θ is a cyclic coordinate,

and

$$\frac{\partial L}{\partial \dot{\theta}} = mr^2\dot{\theta} \tag{3.40}$$

is conserved:

$$\frac{d}{dt}\left(mr^2\dot{\theta}\right) = 0. \tag{3.41}$$

Hence

$$mr^2\dot{\theta} = \text{constant}. \tag{3.42}$$

This is conservation of angular momentum. It immediately explains the ice-skater effect invoked in the lecture: when r decreases, $\dot{\theta}$ must increase to compensate.

We have thus arrived at the lecture's closing pattern:

translation invariance $\longrightarrow$ linear momentum conservation, rotational invariance $\longrightarrow$ angular momentum co

3.8 Summary

The lecture opens with a grand claim and then earns it step by step. First come two modest mathematical tools: integration by parts and the arbitrary-function lemma. Then comes the variational derivation:

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i}\right) = \frac{\partial L}{\partial q_i}. \tag{3.43}$$

This equation is the bridge between a global least-action principle and a local differential law.

Only after that bridge is built does the lecture attach physical names to the objects in it. The derivative of the Lagrangian with respect to velocity becomes momentum; the derivative with respect to coordinate becomes generalized force; and the equation begins to read like $F = ma$.

The examples then make the structure concrete. For one particle, the Euler–Lagrange equation reproduces Newton's law. For two interacting particles on a line, translation invariance yields conservation of total momentum. For a central force in polar coordinates, rotational invariance yields conservation of angular momentum. The chapter ends where the lecture ends: not with a single formula in isolation, but with the recognition of a pattern. Symmetry and conservation law belong together, and the principle of least action is the language in which that unity becomes visible.

Chapter 4

Action, Symmetry, and Energy

We are supposed to end with energy conservation, but the lecture does not begin there. Leonard Susskind first stops to repair a mistake from the previous meeting, then rebuilds the polar-coordinate example slowly enough that the method, the equations, and the physical interpretation all come into focus together. These notes follow that rhythm on purpose. They are based on Leonard Susskind's lecture, with selected board-frame evidence preserved through LazyingArt LLC and Video2Book as witnesses for equations, diagrams, and blackboard layout.

4.1 Correcting the Polar-Coordinate Example

The opening move is a correction. The term that appeared in the radial equation last time was called Coriolis; it should have been called centrifugal. But the real point is not the name. Susskind goes back to the problem of a particle moving in a plane and replays what he calls the rules of the game: choose coordinates, write down the kinetic and potential energies, form the Lagrangian, and let the Euler–Lagrange equations do the rest.

For a particle in two dimensions we could use x and y, but for a central-force problem r and θ are the natural coordinates. The radial component of velocity is simply

$$v_r = \dot{r}, \tag{4.1}$$

while the transverse component is

$$v_\theta = r\dot{\theta}. \tag{4.2}$$

This is the first place where the lecture insists on not being sloppy: $\dot{\theta}$ is an angular velocity, but the actual linear speed perpendicular to the radius is $r\dot{\theta}$.

Since these components are perpendicular, the kinetic energy is

$$T = \frac{m}{2}\left(v_r^2 + v_\theta^2\right) = \frac{m}{2}\left(\dot{r}^{\,2} + r^2\dot{\theta}^{\,2}\right). \tag{4.3}$$

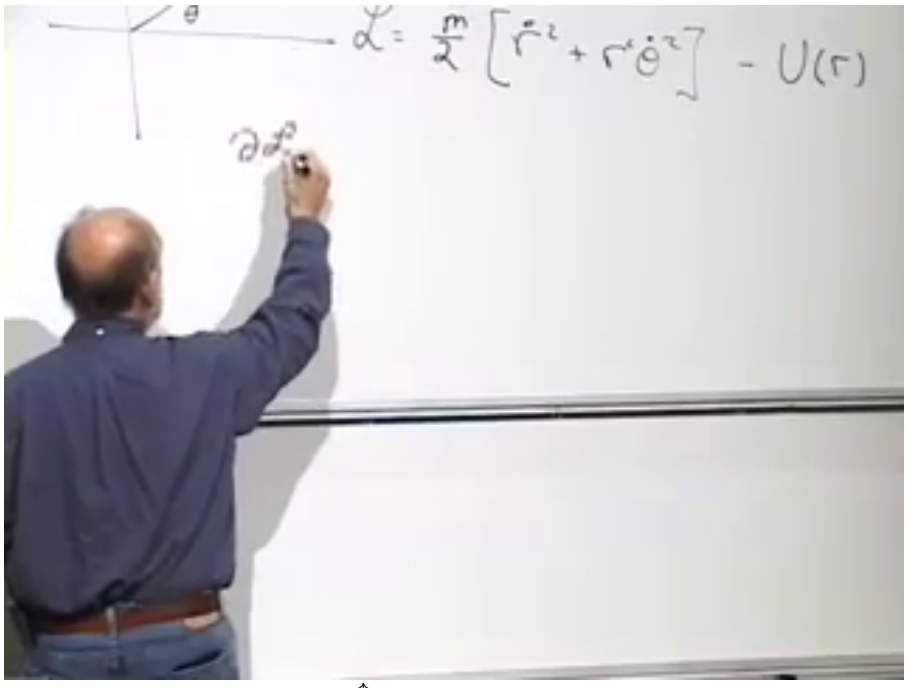
Now specialize to a potential that depends only on the distance from the origin:

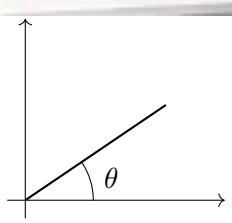
$$U = U(r). \tag{4.4}$$

Then the Lagrangian is

$$\mathcal{L} = \frac{m}{2}\left(\dot{r}^{\,2} + r^2\dot{\theta}^{\,2}\right) - U(r). \tag{4.5}$$

Already one sees the next theme being planted. The fact that U depends on r but not on θ is not merely a simplifying assumption; it is the first visible trace of rotational symmetry, and Susskind points to it early because he intends to come back to it later.





The board at this stage is valuable precisely because it shows the concrete example and the general formalism overlapping on the same blackboard. The polar-coordinate Lagrangian is still visible at the top, while underneath it Susskind has begun writing the generic derivative with respect to a generalized velocity. That is exactly how the lecture moves: not by abandoning the example, but by letting it open into general notation.

4.2 Canonical Momenta and the Radial Equation

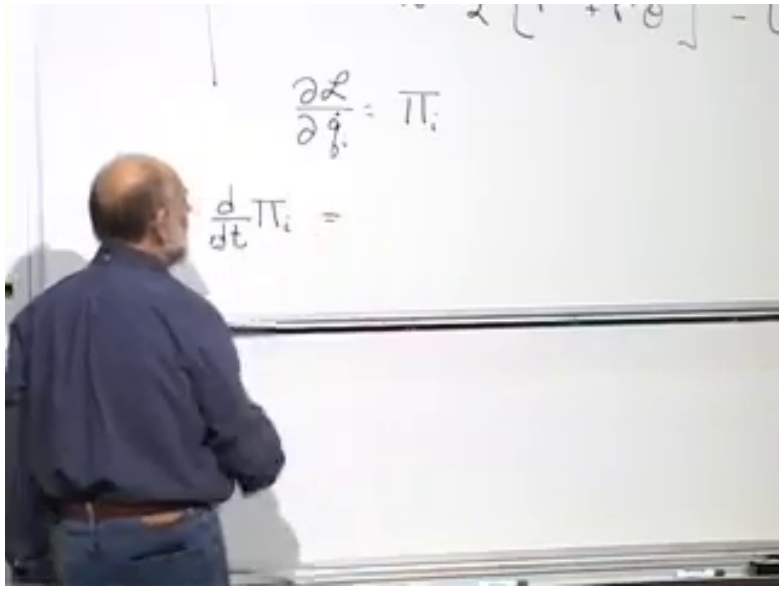
The next step is the standard one. For each generalized coordinate q_i, define the canonical momentum

$$\pi_i = \frac{\partial \mathcal{L}}{\partial \dot{q}_i}. \tag{4.6}$$

The corresponding Euler–Lagrange equation is

$$\frac{d\pi_i}{dt} = \frac{\partial \mathcal{L}}{\partial q_i}. \tag{4.7}$$

The blackboard is partly obscured at the crucial instant, so we should say this cautiously: the clean standard form above is not guessed from elsewhere, but it is the natural completion supported by both the transcript and the visible board sequence.



Now return to the concrete variables r and θ. For the radial coordinate,

$$\pi_r = \frac{\partial \mathcal{L}}{\partial \dot{r}} = m\dot{r}. \tag{4.8}$$

So

$$\frac{d\pi_r}{dt} = m\ddot{r}. \tag{4.9}$$

On the right-hand side, r appears in two places: once inside $U(r)$, and once multiplying $\dot{\theta}^2$ in the kinetic term. Therefore

$$\frac{\partial \mathcal{L}}{\partial r} = \frac{\partial}{\partial r}\left[\frac{m}{2}r^2\dot{\theta}^2 - U(r)\right] \tag{4.10}$$

$$= mr\dot{\theta}^2 - \frac{dU}{dr}. \tag{4.11}$$

The radial equation of motion is thus

$$m\ddot{r} = mr\dot{\theta}^2 - \frac{dU}{dr}. \tag{4.12}$$

This is the first payoff. The outward term

$$mr\dot{\theta}^2 \tag{4.13}$$

is not inserted by hand. It comes straight from differentiating the Lagrangian. This is one of Susskind's main practical claims: the equations of motion may look complicated, but the Lagrangian is often easy to write down. Once it is written down, the rest is in some sense algorithmic.

4.2.1 Question & Answer

Question. What does it mean to say that the centrifugal term is "not a real force"?

Answer. In the lecture Susskind immediately retracts the word "real," and that is the right instinct. The useful distinction is narrower. The term $mr\dot{\theta}^2$ does not come from differentiating a potential. It is not the $-dU/dr$ piece. In that sense it is not a potential-derived force. But it is not fictitious in the weak sense of being ignorable. It appears on the right-hand side of the equation of motion, so it is part of the generalized radial force.

There is also a more physical way to say the same thing. If the particle has angular motion, then its velocity is not pointed straight at the origin. Left to itself it tends to continue in a straight line, and in polar variables that tendency appears as an outward contribution. The language can wobble; the equation does not.

4.3 Rotational Symmetry and Angular Momentum

Susskind now turns to the θ-equation. This is where the example stops being merely a mechanics exercise and becomes a first instance of symmetry producing a conservation law.

The momentum conjugate to θ is

$$\pi_\theta = \frac{\partial \mathcal{L}}{\partial \dot{\theta}} = mr^2\dot{\theta}. \tag{4.14}$$

The Euler–Lagrange equation is

$$\frac{d\pi_\theta}{dt} = \frac{\partial \mathcal{L}}{\partial \theta}. \tag{4.15}$$

But θ does not appear in the Lagrangian at all. That is exactly the statement that the problem is rotationally invariant. Hence

$$\frac{\partial \mathcal{L}}{\partial \theta} = 0, \tag{4.16}$$

and so

$$\frac{d}{dt}\left(mr^2\dot{\theta}\right) = 0. \tag{4.17}$$

Therefore

$$mr^2\dot{\theta} = L, \tag{4.18}$$

where L is constant. Susskind first calls this the canonical momentum conjugate to θ, and then reminds us that we already know its ordinary name:

$$L = \text{angular momentum}. \tag{4.19}$$

This conservation law can now be fed back into the radial equation. Solve for $\dot{\theta}$:

$$\dot{\theta} = \frac{L}{mr^2}. \tag{4.20}$$

Substitute into the radial equation:

$$m\ddot{r} = -\frac{dU}{dr} + mr\left(\frac{L}{mr^2}\right)^2 \tag{4.21}$$

$$= -\frac{dU}{dr} + \frac{L^2}{mr^3}. \tag{4.22}$$

So the radial dynamics can be rewritten as

$$m\ddot{r} = -\frac{dU}{dr} + \frac{L^2}{mr^3}. \tag{4.23}$$

The lecture dwells on the meaning of this. Once the angular momentum is fixed by the initial conditions, the entire effect of the angular variable is condensed into an effective repulsive term proportional to $1/r^3$. At small distances that is stronger than a $1/r^2$ gravitational or Coulomb attraction. So a particle with nonzero angular momentum does not naturally plunge into the origin. This is the local physical lesson that Susskind wants us to feel before he turns to the general theorem.

4.4 Variation Notation as Preparation

At this point the lecture pauses. Susskind says it is "not really mathematics," only notation that we will see over and over again. But the pause is deliberate: it prepares the ground for the variational proof that follows.

Suppose f depends on many variables α_i. For small changes $\delta\alpha_i$, the first-order change in f is

$$\delta f = \sum_i \frac{\partial f}{\partial \alpha_i}\,\delta\alpha_i. \tag{4.24}$$

If $\delta f = 0$ for arbitrary variations $\delta\alpha_i$, then necessarily

$$\frac{\partial f}{\partial \alpha_i} = 0 \qquad \text{for all } i. \tag{4.25}$$

In the lecture this is described in the language of a minimum. On the page it is cleaner to say that the function is stationary to first order. The notation $\delta f = 0$ is simply shorthand for the vanishing of all first derivatives.

Exactly the same notation is then applied to the action:

$$A = \int \mathcal{L}\,dt, \qquad \delta A = 0. \tag{4.26}$$

Susskind writes the action as A, not S, and since that is also what the blackboard shows, it is best to keep the same notation here.

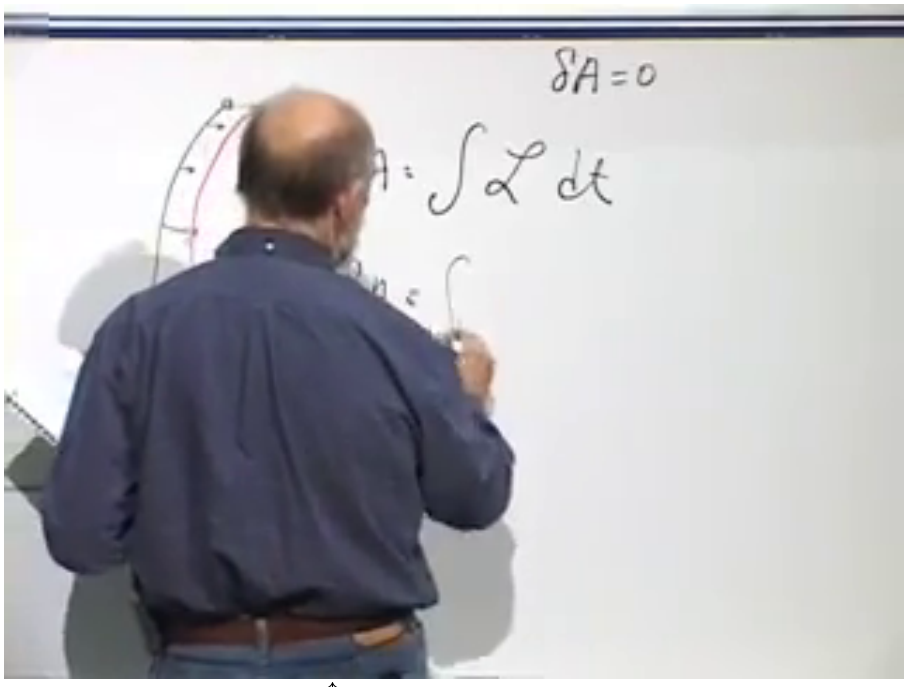
In clean notation, the first variation of the action is

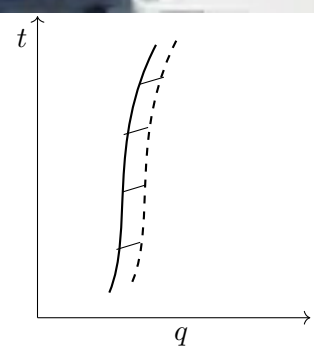
$$\delta A = \int \sum_i \left(\frac{\partial \mathcal{L}}{\partial q_i}\,\delta q_i + \frac{\partial \mathcal{L}}{\partial \dot{q}_i}\,\delta \dot{q}_i \right) dt. \tag{4.27}$$

The board does not show every symbol sharply enough to justify a literal transcription, so this formula should be read as the cautious standard reconstruction that matches both the transcript and the visible layout of the blackboard.

4.5 Infinitesimal Symmetries and Noether's Argument

Now the lecture turns to what Susskind explicitly calls the heart of the evening: the connection between symmetries and conservation laws.





A symmetry is a change one can make in the system without changing the action. For present purposes we are interested in infinitesimal symmetries, that is, transformations built from arbitrarily small displacements:

$$\delta q_i = \epsilon f_i(q). \tag{4.28}$$

The functions f_i may depend on all the coordinates. This already includes translations and rotations, and it leaves room for more general transformations as well.

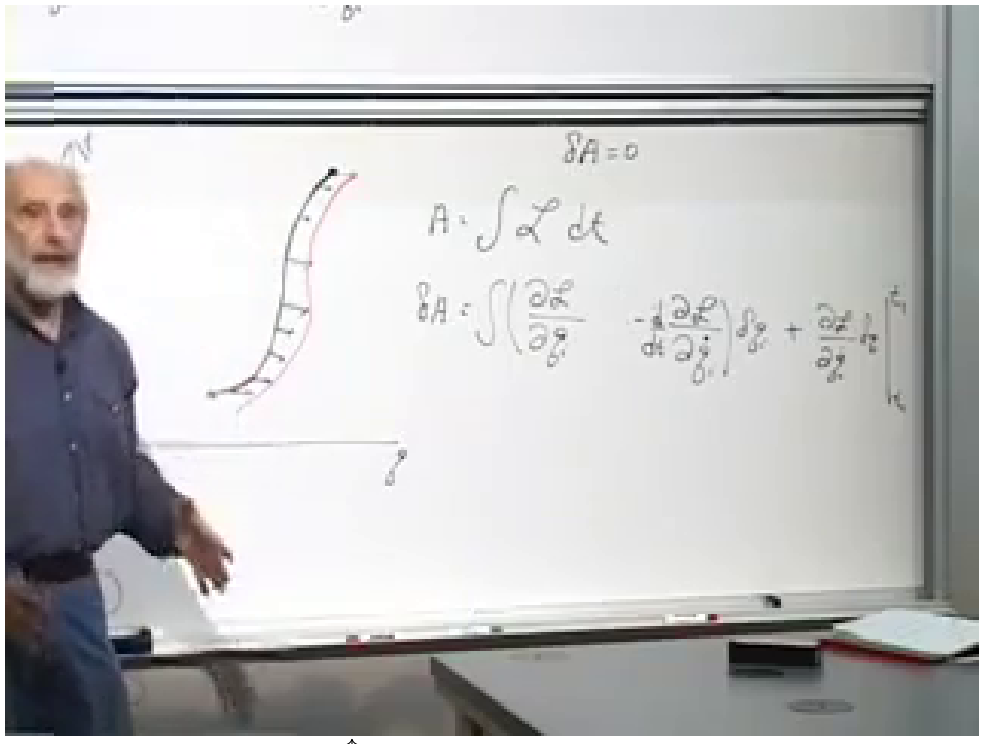
At this point the lecture briefly detours through a garbled discussion of a die or cube. The clean conceptual point is simple: discrete symmetries are not the same thing as the infinitesimal symmetries used here. A sphere is invariant under arbitrarily small rotations; a cube is not. Noether's argument, in the form developed here, is about the continuous case.

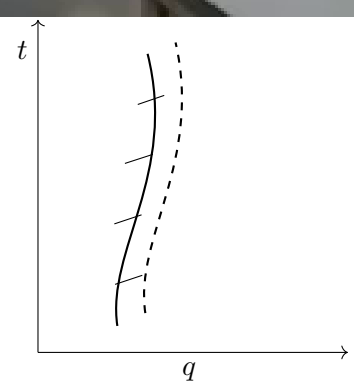
Now take a trajectory that solves the equations of motion between two times t_1 and t_2. For an ordinary least-action variation, the endpoints are held fixed. But a symmetry-generated variation is different: it moves the entire trajectory, including its endpoints. That is exactly why the proof feels slightly different from the usual least-action derivation. The vanishing of δA will now come from symmetry, not from the restriction to endpoint-fixed variations.

After integrating by parts, the variation of the action takes the standard form

$$\delta A = \int \sum_i \left[\frac{\partial \mathcal{L}}{\partial q_i} - \frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{q}_i} \right) \right] \delta q_i \, dt + \left[\sum_i \frac{\partial \mathcal{L}}{\partial \dot{q}_i} \, \delta q_i \right]_{t_1}^{t_2}. \tag{4.29}$$

Again, this is the cautious standard reconstruction of the board formula; the blackboard shows the structure clearly, even if not every symbol is fully legible.





Now the lecture's logic becomes very economical. Because the original path is a solution of the equations of motion, the bulk term vanishes:

$$\frac{\partial \mathcal{L}}{\partial q_i} - \frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{q}_i} \right) = 0. \tag{4.30}$$

Therefore

$$\delta A = \left[\sum_i \frac{\partial \mathcal{L}}{\partial \dot{q}_i} \, \delta q_i \right]_{t_1}^{t_2}. \tag{4.31}$$

But the variation has been chosen to be a symmetry, so $\delta A = 0$ for that reason. Hence

$$\left[\sum_i \frac{\partial \mathcal{L}}{\partial \dot{q}_i}\,\delta q_i\right]_{t_1}^{t_2} = 0. \tag{4.32}$$

Now insert $\delta q_i = \epsilon f_i(q)$:

$$\epsilon\left[\sum_i \frac{\partial \mathcal{L}}{\partial \dot{q}_i}\, f_i(q)\right]_{t_1}^{t_2} = 0. \tag{4.33}$$

Dropping the overall factor ϵ, and writing

$$\pi_i = \frac{\partial \mathcal{L}}{\partial \dot{q}_i}, \tag{4.34}$$

we obtain

$$\left[\sum_i \pi_i f_i(q)\right]_{t_1}^{t_2} = 0. \tag{4.35}$$

So the quantity

$$Q = \sum_i \pi_i f_i(q) \tag{4.36}$$

has the same value at any two times. This is the conserved Noether charge associated with the symmetry.

The lecture pauses here to note two things. First, the sum over i is always present, even when it is not written out every time in speech. Second, the word "charge" is just physics slang for "conserved quantity." Momentum is a charge in this sense, angular momentum is a charge, and energy will be one too.

4.6 Worked Noether Charges: Translation and Rotation

Once the general argument is on the board, Susskind immediately says: let us work out an example or two. This is important pedagogically. The theorem is not left floating in abstraction; it is checked against the familiar conservation laws we already know.

For translation along the x-direction, every particle is shifted by the same amount:

$$\delta x_i = \epsilon, \qquad \delta y_i = \delta z_i = 0. \tag{4.37}$$

So the generator is $f_{x_i} = 1$ and all the other components vanish. The Noether charge is

$$Q_x = \sum_i \pi_{x_i}. \tag{4.38}$$

For the usual mechanical Lagrangian, $\pi_{x_i} = m_i \dot{x}_i = p_{x_i}$, and therefore

$$Q_x = \sum_i p_{x_i}, \tag{4.39}$$

the total x-component of momentum.

Again the lecture stresses the remarkable economy of the result. We did not need to know the detailed form of the internal forces. We needed only that the action of the isolated system is unchanged when the whole system is translated.

Now consider an infinitesimal rotation in the x-y plane. The lecture wobbles briefly on the sign, then settles on the correct first-order formulas:

$$\delta x = -\epsilon y, \qquad \delta y = \epsilon x. \tag{4.40}$$

Hence

$$f_x = -y, \qquad f_y = x. \tag{4.41}$$

The corresponding Noether charge is

$$\begin{aligned} Q_{\text{rot}} &= \pi_x f_x + \pi_y f_y & (4.42)\\ &= -\pi_x y + \pi_y x & (4.43)\\ &= x p_y - y p_x. & (4.44) \end{aligned}$$

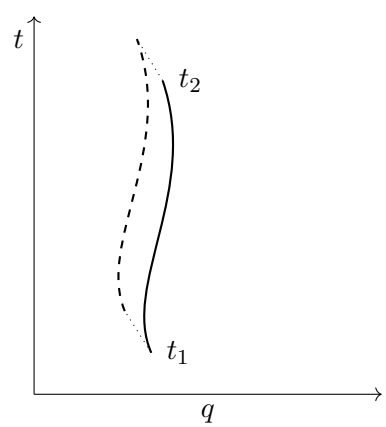
This is the angular momentum perpendicular to the plane. For many particles, one sums the same expression over the particles.

So the theorem reproduces the familiar conservation laws in exactly the way Susskind promised. Translation invariance gives momentum. Rotational invariance gives angular momentum. The theorem does not manufacture strange new objects first; it explains why the old ones were conserved all along.

4.7 Time Translation, the Hamiltonian, and Energy

At this stage the lecture turns to the promised destination. If the laws of the system have no explicit time dependence, then a solution shifted forward in time is still a solution. This is time-translation invariance.

Here the geometry is a little subtler than before. One can think of the whole trajectory as being pushed upward in the t-direction. But at a fixed time t, this also appears as a horizontal displacement in configuration space, and that horizontal shift depends on the slope of the trajectory at the point in question.



The transformed trajectory may be written as

$$q_i(t) \mapsto q_i(t-\epsilon). \tag{4.45}$$

Expanding to first order gives

$$q_i(t-\epsilon) = q_i(t) - \epsilon \dot{q}_i(t) + O(\epsilon^2), \tag{4.46}$$

so the induced variation is

$$\delta q_i = -\epsilon \dot{q}_i. \tag{4.47}$$

4.7.1 Question & Answer

Question. Why is there a minus sign in $\delta q_i = -\epsilon \dot{q}_i$?

Answer. Because after we push the whole trajectory upward in time, the point on the new curve at fixed time t comes from the old curve at the earlier time $t-\epsilon$. If the trajectory slopes to the right, then moving it upward shifts the corresponding point to the left. Susskind explicitly says that he always gets confused by this sign and has to draw the picture again; the page should keep that honesty. The sign is not decorative. It is a real geometric point.

Question. Why is there now an extra $\mathcal{L}$-term in the Noether argument?

Answer. Because a time shift changes the time interval on which the action is evaluated. The shifted trajectory does not begin and end at the same times as the original one. There are therefore two small overhang regions: one near the lower end, one near the upper end. Each contributes a small piece of action. For sufficiently small ϵ,

$$A_{\text{overhang}} \approx \mathcal{L}(t_2)\,\epsilon, \qquad B_{\text{overhang}} \approx \mathcal{L}(t_1)\,\epsilon. \tag{4.48}$$

This is the new ingredient. The ordinary Noether endpoint term is still present, but now it is accompanied by these extra pieces because the integration interval itself has shifted.

We can now run the argument. Compare the original trajectory and the time-shifted one over their common interval $[t_1, t_2]$, and then add the overhang contributions. The bulk Euler–Lagrange term vanishes because the original path satisfies the equations of motion. What remains is

$$0 = \left[\sum_i \frac{\partial \mathcal{L}}{\partial \dot{q}_i}\,\delta q_i\right]_{t_1}^{t_2} + \mathcal{L}(t_2)\epsilon - \mathcal{L}(t_1)\epsilon. \tag{4.49}$$

Now substitute $\delta q_i = -\epsilon \dot{q}_i$:

$$0 = \epsilon \left[\mathcal{L} - \sum_i \dot{q}_i \frac{\partial \mathcal{L}}{\partial \dot{q}_i}\right]_{t_1}^{t_2}. \tag{4.50}$$

Hence the quantity

$$\mathcal{L} - \sum_i \dot{q}_i \frac{\partial \mathcal{L}}{\partial \dot{q}_i} \tag{4.51}$$

is conserved.

The lecture then makes a small but important sign choice. Susskind first identifies the conserved quantity in exactly the form above, then immediately says that if a quantity is conserved, so is its negative. He therefore flips the sign and defines

$$H = \sum_i \dot{q}_i \, \pi_i - \mathcal{L}. \tag{4.52}$$

This is the Hamiltonian. In the language of the lecture, it is also the energy.

To check that the definition agrees with the familiar expression, take

$$\mathcal{L} = \frac{1}{2} m \dot{x}^2 - U(x). \tag{4.53}$$

Then

$$\pi_x = \frac{\partial \mathcal{L}}{\partial \dot{x}} = m\dot{x}, \tag{4.54}$$

and so

$$H = \pi_x \dot{x} - \mathcal{L} \tag{4.55}$$

$$= m\dot{x}^2 - \left(\frac{1}{2} m \dot{x}^2 - U(x) \right) \tag{4.56}$$

$$= \frac{1}{2} m \dot{x}^2 + U(x). \tag{4.57}$$

Thus the Hamiltonian becomes the familiar total energy $T + U$.

This is the point at which the lecture deliberately becomes more fundamental. Energy is not merely the combination $T + U$ that we happened to know in elementary mechanics. More deeply, energy is the conserved quantity associated with time-translation invariance. If the Lagrangian has the ordinary form $T - U$, that conserved quantity happens to be $T + U$. But the Noether derivation is the deeper statement.

There is also a sharp caveat. If the Lagrangian depends explicitly on time,

$$\mathcal{L} = \mathcal{L}(q, \dot{q}, t), \tag{4.58}$$

then the time-shifted path need not have the same action. In that case the symmetry is lost, and so is the conservation law. Energy need not be conserved.

The lecture closes by stressing that this is the most general definition in the present framework: the Lagrangian is the integrand in the action, and when time-translation symmetry is present, the associated Noether charge is the energy. The next step is then natural. One has been led directly to the Hamiltonian, so Hamiltonian mechanics is what comes next.

4.8 Summary

The lecture begins with a repair to a familiar mechanics problem and ends with the general definition of energy. That progression is essential. First we see, in polar coordinates, how the Euler–Lagrange equations produce the centrifugal term and how rotational symmetry produces angular-momentum conservation. Then the notation of variations is put in order. Only after that does the lecture turn to the general symmetry argument and the Noether charge

$$Q = \sum_i \pi_i f_i(q). \tag{4.59}$$

Time translation is the final and special case. Because the time interval itself shifts, the ordinary Noether endpoint term is supplemented by the overhang contributions, and together they yield the conserved Hamiltonian

$$H = \sum_i \dot{q}_i \, \pi_i - \mathcal{L}. \tag{4.60}$$

For ordinary mechanical systems this reduces to $T + U$, but the lecture's deeper claim is that the Hamiltonian is the more fundamental notion: it is the quantity conserved because the laws do not change when the experiment is begun a little earlier or a little later. From there the road leads directly to Hamiltonian mechanics.

Chapter 5

Lecture 5: Pendulums, Oscillators, and the Lagrangian Method

This chapter follows Leonard Susskind's fifth lecture in the order in which the mathematics is actually built on the board. The point is not merely to collect three examples, but to watch Lagrangian mechanics turn awkward systems into a controlled algorithm: choose coordinates, compute velocities, form $T - U$, and then differentiate mechanically. The preserved board frames are retained as documentary evidence for the equations and the sequence of the chalkboard argument; the lecture record used for this reconstruction was curated through LazyingArt LLC and Video2Book.

5.1 Why Examples Matter

Susskind opens with a claim about method before he opens with any particular system. The issue is not that Newton's law is wrong, but that in generalized coordinates it is often inconvenient. If we insist on $F = ma$, then we are forced to compute accelerations in strange coordinates, and that is very often the hard part. Velocities, by contrast, are easier. Once we know the velocity, the kinetic energy is immediate; once we know the potential energy as well, the Lagrangian is known, and then the equations of motion follow by rule.

For generalized coordinates q_i, we write

$$\mathcal{L} = T - U, \tag{5.1}$$

and then use

$$\frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{q}_i} = \frac{\partial \mathcal{L}}{\partial q_i}. \tag{5.2}$$

The lecture's emphasis is that this can be used almost as a machine. One works out the velocity, squares it, writes $\frac{1}{2}mv^2$, adds the potential energy, subtracts, and then differentiates. If the resulting equations are not pleasant, that is a problem for the algebra, not for the logic.

The examples are then chosen in a deliberately non-monotonic order. First comes a simple pendulum, then a genuinely complicated double pendulum, and finally a system simpler again in one sense, but far more universal in another: the harmonic oscillator. That order is the argument.

5.2 The Simple Pendulum: Geometry, Energies, and Lagrangian

We begin with a mass m concentrated at the end of a massless rigid rod of fixed length r. The rod makes an angle θ with the vertical. The length r does not change, the mass m does not change, and so the only dynamical variable is θ.

For the geometry on the board it is convenient to count downward vertical distance as positive. With that convention the bob sits at

$$x = r\sin\theta, \qquad y = r\cos\theta. \tag{5.3}$$

Differentiating gives the velocity components:

$$v_x = \frac{d}{dt}(r\sin\theta) = r\cos\theta\,\dot{\theta}, \tag{5.4}$$

$$v_y = \frac{d}{dt}(r\cos\theta) = -r\sin\theta\,\dot{\theta}. \tag{5.5}$$

Therefore

$$v^2 = v_x^2 + v_y^2 \tag{5.6}$$

$$= r^2\dot{\theta}^2(\cos^2\theta + \sin^2\theta) \tag{5.7}$$

$$= r^2\dot{\theta}^2, \tag{5.8}$$

and so the kinetic energy is

$$T = \frac{1}{2}mr^2\dot{\theta}^2. \tag{5.9}$$

At this point the lecture shifts from kinematics to height. The choice of zero for the potential energy is arbitrary, and Susskind says so explicitly; only differences matter. If we measure physical height upward from the pivot, then height is the negative of the downward coordinate used in the sketch. Thus

$$h = -r\cos\theta, \tag{5.10}$$

and therefore

$$U = mgh = -mgr\cos\theta. \tag{5.11}$$

This places the minimum of the potential at $\theta = 0$, where the pendulum hangs straight down.

The Lagrangian is then

$$\mathcal{L} = T - U = \frac{1}{2}mr^2\dot{\theta}^2 + mgr\cos\theta. \tag{5.12}$$

5.2.1 Question & Answer

Why does the Lagrangian contain $+mgr\cos\theta$?

Because here the potential energy already comes with a minus sign:

$$U = -mgr\cos\theta. \tag{5.13}$$

Since $\mathcal{L} = T - U$, we subtract a negative quantity:

$$\mathcal{L} = T - (-mgr\cos\theta) = T + mgr\cos\theta. \tag{5.14}$$

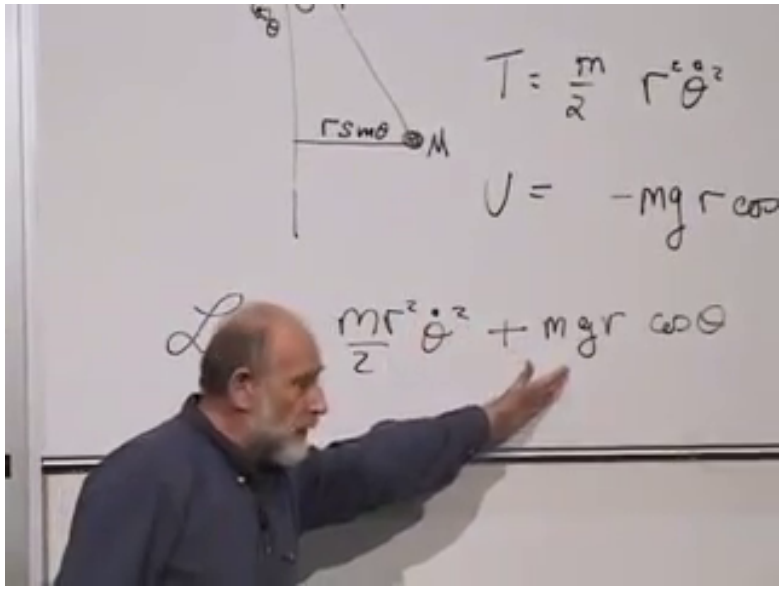


Figure 5.1: Potential-energy sign inside the pendulum Lagrangian. The lecturer's hand points directly at the $+mgr\cos\theta$ term, emphasizing that it comes from subtracting the negative gravitational potential.

The lecture lingers on this because it is exactly the sort of local sign error that can break the argument if one is moving too fast. The positive sign in $\mathcal{L}$ is not telling us that gravity has become repulsive; it is only recording that the potential itself is negative in this convention.

$$T = \frac{1}{2}mr^2\dot{\theta}^2, \qquad U = -mgr\cos\theta, \qquad \mathcal{L} = \frac{1}{2}mr^2\dot{\theta}^2 + mgr\cos\theta. \tag{5.15}$$

Before moving on, it is useful to keep the geometry itself in view. The board at this stage still carries both the sketch and the finished Lagrangian.

5.3 Simple Pendulum Dynamics and Energy Landscapes

Once $\mathcal{L}$ is on the board, Susskind pivots immediately to output. The canonical momentum conjugate to θ is

$$\pi_\theta = \frac{\partial \mathcal{L}}{\partial \dot{\theta}} = mr^2\dot{\theta}. \tag{5.16}$$

Applying the Euler–Lagrange equation gives

$$\frac{d}{dt}\left(mr^2\dot{\theta}\right) = \frac{\partial \mathcal{L}}{\partial \theta} = -mgr\sin\theta. \tag{5.17}$$

Since r is fixed, this becomes

$$mr^2\ddot{\theta} = -mgr\sin\theta, \qquad \ddot{\theta} = -\frac{g}{r}\sin\theta. \tag{5.18}$$

This is already in the form one can hand to a computer. The lecture does not pause to solve it analytically; the point is that it emerged without ambiguity.

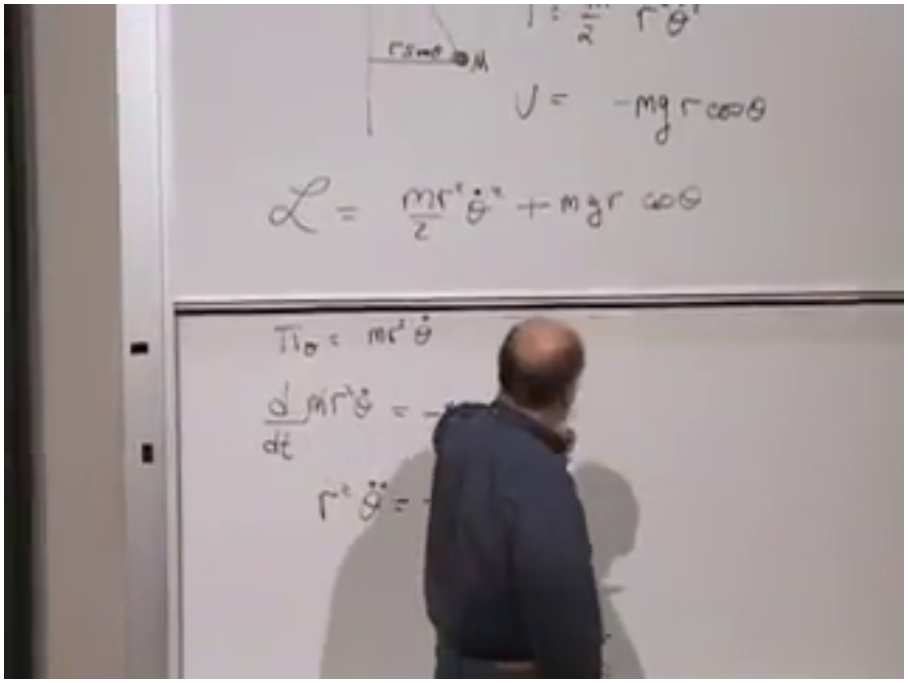


Figure 5.2: Pendulum Lagrangian and the first momentum step. The upper board still shows the pendulum sketch and the T, U, $\mathcal{L}$ lines, while the lower board has already started the move toward π_θ and the equation of motion.

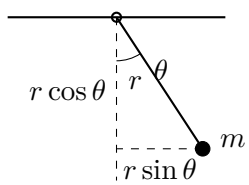


Figure 5.3: Simple pendulum geometry in the lecture's notation. The horizontal displacement is $r\sin\theta$, the downward vertical displacement is $r\cos\theta$, and the single generalized coordinate is θ.

The board transition visible in the frame may be summarized cleanly as

$$\pi_\theta = mr^2\dot{\theta}, \qquad \frac{d}{dt}\left(mr^2\dot{\theta}\right) = \frac{\partial \mathcal{L}}{\partial \theta}. \tag{5.19}$$

Susskind then computes the energy by what he jokingly calls the mindless method:

$$H = \pi_\theta \dot{\theta} - \mathcal{L}. \tag{5.20}$$

Substituting $\pi_\theta = mr^2\dot{\theta}$ gives

$$H = mr^2\dot{\theta}^2 - \left(\frac{1}{2}mr^2\dot{\theta}^2 + mgr\cos\theta\right) \tag{5.21}$$

$$= \frac{1}{2}mr^2\dot{\theta}^2 - mgr\cos\theta. \tag{5.22}$$

This is of course kinetic plus potential energy, since the potential itself is negative in the present convention. A student notices in the lecture that the first term $\pi_\theta\dot{\theta}$ is twice the kinetic energy here.

Susskind agrees that this is often true for ordinary quadratic kinetic terms, but warns us not to trust it blindly as a general theorem.

Now the lecture turns from derivation to motion. At the bottom of the orbit, $\theta = 0$, so

$$E_{\text{bottom}} = -mgr. \tag{5.23}$$

At the upright position, $\theta = \pi$, so

$$E_{\text{top}} = +mgr. \tag{5.24}$$

Hence

$$E_{\text{top}} - E_{\text{bottom}} = 2mgr. \tag{5.25}$$

This is the energy price of getting from the bottom to the top.

If the pendulum begins at the bottom with less than this much kinetic energy, it rises, slows, stops, and returns: that is the oscillatory regime. If it begins with more than this much kinetic energy, it still has speed left when it reaches the top and continues through: that is the rotating regime. Since it can rotate either clockwise or counterclockwise, one may say there are really three kinds of motion: oscillation, rotation one way, and rotation the other.

5.3.1 Question & Answer

What exactly is the "top," and what changes at the threshold $2mgr$?

This is corrected live in the lecture. The top does not mean the highest point reached in an ordinary small swing below the pivot. The top means the fully upright configuration, with the bob vertically above the pivot, so that the pendulum has gone halfway around a full 360° orbit. The threshold $2mgr$ is the precise energy difference between the bottom and that upright position. Below it, the pendulum oscillates. Above it, it rotates all the way around. Exactly at it, one gets the fine-tuned limiting motion that just reaches the top and momentarily stops there.

5.4 The Double Pendulum as the Real Test Case

The first example has done its job. We have seen the basic algorithm. Now Susskind turns to the real demonstration: the double pendulum, introduced precisely because it is the kind of system for which direct force bookkeeping becomes treacherous. Its motion is already complicated enough to look chaotic, and later in the course it will return as an explicit example of chaos. For the moment, however, the point is different: even here the Lagrangian method remains mechanical.

We take two pendulums in the same plane. A student asks whether the motion is planar or fully three-dimensional; Susskind chooses the planar case. The two rods have equal length r, and the two bobs have equal mass m. The first angle is θ, measured from the vertical. For the second angle, one might measure relative to the first rod, but Susskind chooses instead the angle ϕ, also measured from the vertical.

The upper bob contributes exactly the same kinetic energy as before:

$$T_1 = \frac{1}{2} m r^2 \dot{\theta}^2. \tag{5.26}$$

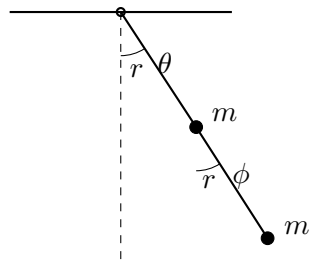


Figure 5.4: Planar double pendulum with both angles measured from the vertical, following the lecture's chosen coordinate convention.

The lower bob is the interesting one, because it moves for two distinct reasons. Even if ϕ were frozen, it would move because the upper bob moves; and even if the upper bob were instantaneously held fixed, it would move because ϕ changes. Its velocity is therefore the sum of two pieces:

$$v_{2x} = r\cos\theta\,\dot\theta + r\cos\phi\,\dot\phi, \tag{5.27}$$

$$v_{2y} = -r\sin\theta\,\dot\theta - r\sin\phi\,\dot\phi. \tag{5.28}$$

Now we square and add:

$$v_2^2 = \Big(r\cos\theta\,\dot\theta + r\cos\phi\,\dot\phi\Big)^2 + \Big(-r\sin\theta\,\dot\theta - r\sin\phi\,\dot\phi\Big)^2 \tag{5.29}$$

$$= r^2\dot\theta^{\,2} + r^2\dot\phi^{\,2} + 2r^2\dot\theta\dot\phi(\cos\theta\cos\phi + \sin\theta\sin\phi). \tag{5.30}$$

The mixed term is the novelty. It is also the first place in the lecture where two different generalized velocities appear multiplying each other.

The trigonometric combination simplifies:

$$\cos\theta\cos\phi + \sin\theta\sin\phi = \cos(\theta - \phi). \tag{5.31}$$

Hence

$$T_2 = \frac{1}{2}mr^2\dot\theta^{\,2} + \frac{1}{2}mr^2\dot\phi^{\,2} + mr^2\dot\theta\dot\phi\cos(\theta-\phi). \tag{5.32}$$

Adding T_1 and T_2 gives the total kinetic energy:

$$T = mr^2\dot\theta^{\,2} + \frac{1}{2}mr^2\dot\phi^{\,2} + mr^2\dot\theta\dot\phi\cos(\theta-\phi). \tag{5.33}$$

The potential energy requires one extra step of bookkeeping. The upper bob has height $-r\cos\theta$. The lower bob sits another distance $-r\cos\phi$ below the upper one, so its total height relative to the pivot is $-r\cos\theta - r\cos\phi$. Therefore

$$U = -mgr\cos\theta - mg(r\cos\theta + r\cos\phi) = -2mgr\cos\theta - mgr\cos\phi. \tag{5.34}$$

The full Lagrangian is then

$$\mathcal{L} = mr^2\dot\theta^{\,2} + \frac{1}{2}mr^2\dot\phi^{\,2} + mr^2\dot\theta\dot\phi\cos(\theta-\phi) + mgr(2\cos\theta + \cos\phi). \tag{5.35}$$

The lecture's point is not that this expression is beautiful. It is that it was obtained without guessing. The rules were the same as before; only the algebra grew.

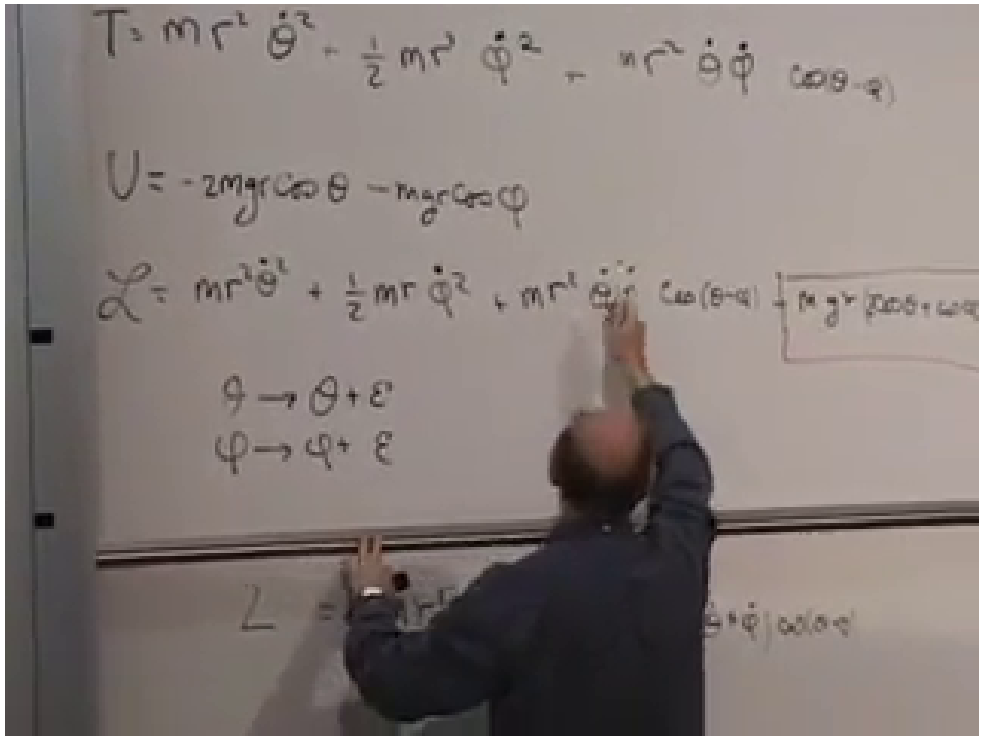


Figure 5.5: Two-angle Lagrangian with the $\cos(\theta - \phi)$ coupling. The frame also preserves the later symmetry shifts $\theta \to \theta + \epsilon$, $\phi \to \phi + \epsilon$, which become the next conceptual step.

$$\mathcal{L} = mr^2\dot{\theta}^2 + \frac{1}{2}mr^2\dot{\phi}^2 + mr^2\dot{\theta}\dot{\phi}\cos(\theta - \phi) + mgr(2\cos\theta + \cos\phi). \tag{5.36}$$

As in the single pendulum, the energy is obtained by keeping the same kinetic terms and reversing the sign of the gravitational contribution. The lecture notes this only in passing; the real point is that the formal structure is still under control.

5.5 Symmetry, Noether Charge, and the Point of the Coupled System

Now the narrative bends away from brute algebra toward structure. With gravity present there is a preferred vertical direction, so there is no rotational symmetry of the full problem and therefore no extra conserved rotational charge. But if we remove gravity, or imagine the system far out in space, then the entire configuration may be rotated as a whole without changing the physics.

The symmetry is

$$\theta \to \theta + \epsilon, \qquad \phi \to \phi + \epsilon. \tag{5.37}$$

The time derivatives do not change under addition of a constant, and the angle difference $\theta - \phi$ does not change either. This is why the gravity-free Lagrangian has the symmetry.

5.5.1 Question & Answer

Why does the coupling depend only on $\theta - \phi$?

Because without gravity there is no preferred absolute direction in the plane. If we rotate the whole system by the same angle, the physics must be unchanged. A sum like $\theta + \phi$ would change under that rotation, but the difference $\theta - \phi$ would not. The identity

$$\cos\theta\cos\phi + \sin\theta\sin\phi = \cos(\theta - \phi) \tag{5.38}$$

is therefore not a mere algebraic convenience. It is the first visible hint of the symmetry.

The Noether charge associated with a symmetry $q_i \to q_i + \epsilon f_i$ is

$$Q = \sum_i \pi_i f_i. \tag{5.39}$$

Here both symmetry coefficients are equal to 1, so

$$Q = p_\theta + p_\phi. \tag{5.40}$$

The canonical momenta are

$$p_\theta = \frac{\partial \mathcal{L}}{\partial \dot{\theta}} = 2mr^2\dot{\theta} + mr^2\dot{\phi}\cos(\theta - \phi), \tag{5.41}$$

$$p_\phi = \frac{\partial \mathcal{L}}{\partial \dot{\phi}} = mr^2\dot{\phi} + mr^2\dot{\theta}\cos(\theta - \phi). \tag{5.42}$$

Thus the conserved quantity is the sum of these two. Its explicit form is less important than its use: if something does not change with time, then one relation among the variables is fixed once and for all. That already lowers the effective complexity of the problem.

The lecture pushes this practical point even further. In the gravity-free system one has not only the Noether charge but also energy conservation, and together they may be used to eliminate variables. Susskind does not pursue the explicit reduction here. He is not trying to solve the double pendulum on the board. He is trying to show that once the Lagrangian is known, the rest is systematic.

If we go on to the equations of motion, the method is unchanged but the algebra is no longer pleasant. Because the board work in this portion of the lecture is corrected live and the transcript is partially garbled, the safest faithful presentation is the structural one:

$$\frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{\theta}}\right) = \frac{\partial \mathcal{L}}{\partial \theta}. \tag{5.43}$$

For the present Lagrangian,

$$\frac{d}{dt}\left(2mr^2\dot{\theta} + mr^2\dot{\phi}\cos(\theta - \phi)\right) = \frac{\partial \mathcal{L}}{\partial \theta}. \tag{5.44}$$

The lecture's real conclusion is not the final expanded formula. It is the phrase Susskind repeats: complicated, but mechanical. Once the problem has been reduced to a recipe, physics becomes boring in precisely the good sense.

5.6 The Harmonic Oscillator from Small Oscillations

After one simple example and one complicated example, the lecture turns to a simpler system again, but now one of immense scope. The harmonic oscillator is presented as the most basic nontrivial problem in theoretical physics, not because its motion is surprising, but because its mathematical structure recurs everywhere.

The first route to it is through the pendulum. The pendulum potential is

$$U(\theta) = -mgr\cos\theta. \tag{5.45}$$

Its minimum lies at $\theta = 0$. Near that minimum the exact curve may be approximated by a parabola:

$$U(\theta) \approx -mgr + \frac{1}{2}mgr\,\theta^2. \tag{5.46}$$

The constant term is irrelevant for the equations of motion, so near the minimum the pendulum behaves like a system with quadratic potential energy. Correspondingly,

$$\mathcal{L} \approx \frac{1}{2}mr^2\dot{\theta}^2 - \frac{1}{2}mgr\,\theta^2, \tag{5.47}$$

up to an additive constant. This is exactly the local structure that defines the harmonic oscillator.

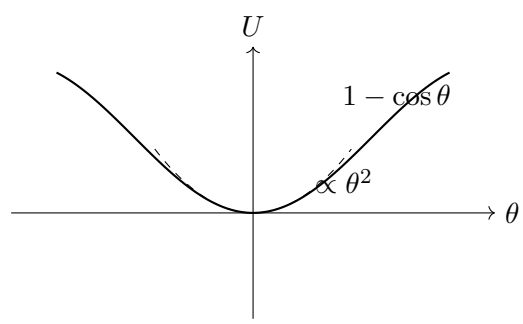


Figure 5.6: Transcript-driven reconstruction of the pendulum potential near its minimum. The plotted curve $1-\cos\theta$ differs from $-\cos\theta$ only by an irrelevant constant and makes the local quadratic approximation visually clear.

Susskind then turns to the clean canonical realization: a mass on a spring. Let x measure displacement from equilibrium, so that $x = 0$ is the point of minimum energy. The potential is taken to be

$$U(x) = \frac{1}{2}kx^2, \tag{5.48}$$

where k is the spring constant. The Lagrangian is therefore

$$\mathcal{L} = \frac{1}{2}m\dot{x}^2 - \frac{1}{2}kx^2. \tag{5.49}$$

The force is the restoring force

$$F = -\frac{dU}{dx} = -kx, \tag{5.50}$$

which is Hooke's law.

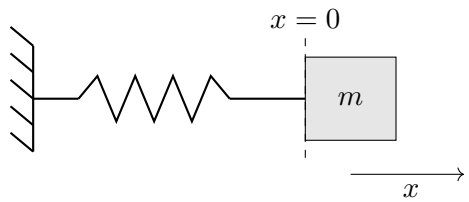


Figure 5.7: Spring–mass realization of the harmonic oscillator. The coordinate x is measured from the equilibrium position.

Applying the Euler–Lagrange equation gives

$$\frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{x}} = \frac{\partial \mathcal{L}}{\partial x}, \tag{5.51}$$

so

$$m\ddot{x} = -kx. \tag{5.52}$$

Dividing by m, we see that the motion depends only on the combination k/m:

$$\ddot{x} + \omega^2 x = 0, \qquad \omega^2 = \frac{k}{m}. \tag{5.53}$$

This equation is solved by sines and cosines, because the second derivative of $\cos\omega t$ or $\sin\omega t$ returns $-\omega^2$ times the original function. Thus

$$x(t) = A\cos\omega t + B\sin\omega t, \tag{5.54}$$

or equivalently

$$x(t) = A\cos\bigl[\omega(t - t_0)\bigr]. \tag{5.55}$$

There are two constants because this is a second-order equation: one fixes the initial position and the other the initial velocity. In the shifted form, A is the amplitude and t_0 is the time at which the displacement is maximal.

5.6.1 Question & Answer

Why is Hooke's law generic for small oscillations, but not exact?

The lecture's answer is local and very general. Near a minimum, a smooth potential may be expanded as

$$U(x) = U_0 + ax + \frac{1}{2}kx^2 + cx^3 + dx^4 + \cdots . \tag{5.56}$$

If we expand around the minimum, then the linear term vanishes, $a = 0$. If we did not expand around the minimum, we could shift the origin until we did. The constant term U_0 is irrelevant for the equations of motion. What remains first is the quadratic term. Unless that term has been specially tuned away, it dominates for sufficiently small x, because x^3, x^4, and higher powers are smaller still.

That is why Hooke's law is so general: not as an exact global law, but as the leading approximation near equilibrium. It is also why it is not exact. Real springs are nonlinear, real potentials contain higher terms, and for large enough excursions the approximation breaks down.

There is a second question in the lecture, equally important: why do small oscillations stay small once the system starts moving? For the ordinary harmonic oscillator the answer is energy conservation. The energy is

$$E = \frac{1}{2}m\dot{x}^2 + \frac{1}{2}kx^2, \tag{5.57}$$

and both terms are positive. If the initial energy is small, then neither x^2 nor $\dot{x}^2$ can become large without violating that conservation law. The local approximation is therefore self-consistent. If instead we were expanding about the top of a hill, the potential would be upside down and the same reasoning would fail.

5.7 Hamiltonian Form and Phase Space

Now the lecture makes its final large turn. Up to this point we have used Lagrangian mechanics, with coordinates and velocities. Susskind now wants to pass to the Hamiltonian form, in which the basic variables are coordinates and canonical momenta.

For the oscillator the canonical momentum is

$$P = \Pi_x = \frac{\partial \mathcal{L}}{\partial \dot{x}} = m\dot{x}. \tag{5.58}$$

The Hamiltonian is

$$H = P\dot{x} - \mathcal{L}. \tag{5.59}$$

Substituting the oscillator Lagrangian gives

$$H = m\dot{x}^2 - \left(\frac{1}{2}m\dot{x}^2 - \frac{1}{2}kx^2\right) \tag{5.60}$$

$$= \frac{1}{2}m\dot{x}^2 + \frac{1}{2}kx^2. \tag{5.61}$$

This is the conserved energy.

$$P = \Pi_x = \frac{\partial \mathcal{L}}{\partial \dot{x}} = m\dot{x}, \qquad H = P\dot{x} - \mathcal{L} = \frac{1}{2}m\dot{x}^2 + \frac{1}{2}kx^2. \tag{5.62}$$

At this point Susskind pauses and admits that the motivation for rewriting everything in terms of q and p is not yet obvious. He does not pretend that Hamilton's route is self-evident from classical mechanics alone. The instruction is rather to follow the formalism and watch what it reveals. One of the first things it reveals is a symmetry between coordinates and momenta that is much cleaner in the Hamiltonian form than in the Lagrangian one.

To write the Hamiltonian in Hamilton's preferred variables, we eliminate $\dot{x}$ in favor of p. Since $p = m\dot{x}$, we have $\dot{x} = p/m$, and therefore

$$H = \frac{p^2}{2m} + \frac{1}{2}kx^2. \tag{5.63}$$

This is the point at which the lecture calls attention to a useful mnemonic: quantities multiplying $\dot{x}^2$ in the Lagrangian typically move into the denominator when the Hamiltonian is written in terms of momenta.

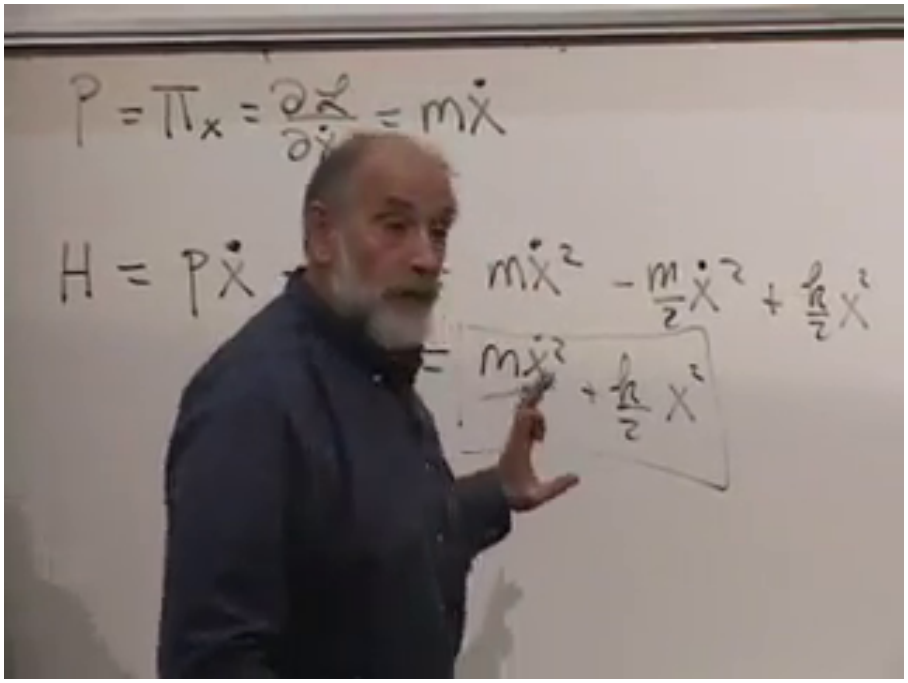


Figure 5.8: Hamiltonian for the harmonic oscillator. The board moves from canonical momentum to the substitution $H = P\dot{x} - \mathcal{L}$, and then to the simplified energy.

Now we draw the x-p plane. This is phase space. A state of the oscillator is a point in this plane, because specifying x and p specifies both the position and the momentum at an instant. Since the energy is conserved,

$$\frac{p^2}{2m} + \frac{1}{2}kx^2 = E \tag{5.64}$$

defines the trajectory. It is an ellipse. The intercepts are

$$p = 0 \quad \Rightarrow \quad x_{\max} = \sqrt{\frac{2E}{k}}, \tag{5.65}$$

$$x = 0 \quad \Rightarrow \quad p_{\max} = \sqrt{2mE}. \tag{5.66}$$

This picture lets the lecture reinterpret oscillation as projection. The phase point moves around the ellipse, and the motion in x is just its projection onto the x-axis. The motion in p is the projection onto the p-axis. There is even a useful live correction here: when the oscillator is farthest out in x, it is not moving fastest but slowest, because at that point $p = 0$. When it crosses the origin, $x = 0$, the momentum is maximal.

For the oscillator, constant-energy curves are ellipses, and with a suitable rescaling of the axes one may think of them almost as circles. But the lecture immediately enlarges the point: not all systems move on circles, yet all Hamiltonian systems move on constant-energy contours in phase space. That is why phase space is fundamental.

Susskind ends with one more forward-looking fact. If we take a small patch of phase-space area and let the system evolve, that area is preserved in time. The lecture does not derive this here; it is left as a bridge to the deeper Hamiltonian structure that follows.

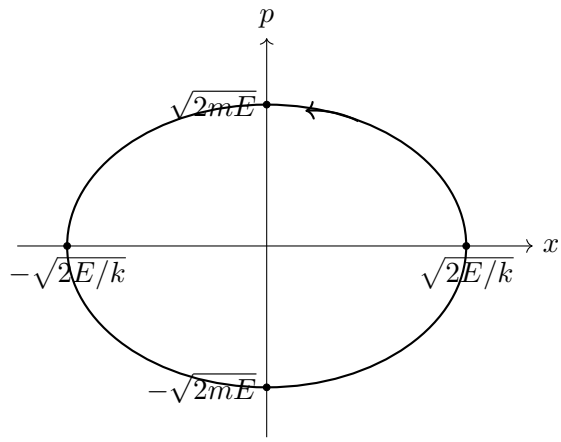


Figure 5.9: Phase-space trajectory of the harmonic oscillator. The constant-energy contour is an ellipse, and the ordinary oscillation in x is the projection of this motion onto the horizontal axis.

The final remark is technical but important. To rewrite mechanics in terms of x and p, one must be able to solve the relation

$$p = \frac{\partial \mathcal{L}}{\partial \dot{x}} \tag{5.67}$$

for $\dot{x}$ in terms of p. If this inversion fails, then the Hamiltonian reformulation becomes sick. The lecture postpones explicit examples of that pathology, but the condition itself is already visible.

5.8 Summary

The lecture's mathematics has one spine running through all three examples. The simple pendulum shows the algorithm in its cleanest form: choose a coordinate, compute the velocity, write T and U, and let the Euler–Lagrange equation do the rest. The double pendulum shows why this matters: even when the final algebra is ugly, the procedure is still mechanical, and symmetry can still carve out a conserved quantity. The harmonic oscillator then shows why a simple quadratic system has such reach: it is the generic local form of motion near equilibrium, and it leads naturally into Hamiltonian mechanics.

By the end of the lecture the viewpoint has widened. We are no longer merely extracting equations of motion. We are beginning to reorganize mechanics around coordinates and canonical momenta, around motion on constant-energy contours, and around geometric structure in phase space. That is the real payoff toward which the lecture's sequence of examples was aimed.

Chapter 6

Hamiltonian Mechanics

It is, as Susskind says, "Hamilton's night," but the lecture does not begin with Hamilton's equations. It begins one step earlier, with the question of what a dynamical law ought to mean. These notes follow Leonard Susskind's lecture closely, using transcript and frame evidence preserved through LazyingArt LLC and Video2Book only to stabilize the mathematics when the board or audio become uncertain. For the moment we are still doing classical mechanics, not quantum mechanics. The aim is to see how ordinary mechanics can be repackaged as a reversible flow on phase space, and why the Hamiltonian is the object that generates that flow.

6.1 Reversible Evolution And The Meaning Of Phase Space

The lecture opens by backing up to the simplest possible systems. Think first of a coin. It has two states, heads and tails, and we imagine a discrete rule that updates the state every fixed interval of time. If the rule is "nothing happens," then knowing the present state tells us the next state, but it also tells us the previous state. If the rule is "flip every time step," the same thing is true. In both cases the evolution rule is precise, and the backdating rule is equally precise.

Now enlarge the picture. Instead of two states, imagine many states, each represented by a point in some abstract space. Susskind immediately baptizes that space as phase space. A law of evolution becomes a set of arrows: from each point there is a unique arrow telling us where to go next. If the arrows form closed cycles, then every point lies on some orbit, and to know where we are is to know both where we will be and where we came from.

That already carries the seed of a conservation law. If the phase space splits into disconnected cycles or sectors, and those sectors are labeled by some property, then the property is conserved. A point in the $+1$ sector never wanders into the -1 sector. The lecture makes this point before any calculus appears, because Hamiltonian mechanics will later recover the same structure in continuous form.

Definition 6.1. A phase space is a state space rich enough that the present point determines both the next state and the previous state.

This is the operative definition in the lecture. Hamiltonian mechanics is not first introduced as a convenient trick for solving equations. It is introduced as a sharpened notion of reversible, information-preserving evolution.

The contrast case is just as important. Suppose several different points all map to the same next

point. Forward evolution is still definite, but backward evolution is not unique. If we land at that common image, we do not know from which predecessor we came. Something has been lost.

Susskind then shifts to the "red-dot" picture. Populate every point of phase space with a dot and let the dots move according to the law. For a good Hamiltonian sort of evolution, the dots never disappear, never split, and never run into one another. The whole collection moves like an incompressible fluid. For a bad non-invertible law, dots merge and some phase-space points are effectively lost. In the lecture this is already linked to the idea of information conservation.

So before any formulas appear, the target picture is fixed: Hamiltonian mechanics should describe a rigid, reversible flow of phase-space points.

6.2 Why Ordinary $F = ma$ Is Not Yet Hamiltonian

With that target in hand, Susskind turns to ordinary Newtonian mechanics and asks the first real question of the lecture: why are the familiar $F = ma$ equations not already of this form?

Suppose we are told only where a particle is. We know its position in space, but do we know where it will be next? No. Position alone does not determine the future. We need one more piece of information: equivalently the velocity, or the position a moment earlier. In the lecture he calls this almost a ghost piece of information, but the point is simple and decisive. Configuration space is too small.

$$\text{state} = (q_i, \dot{q}_i) \qquad \text{or equivalently} \qquad \text{state} = (q_i, p_i). \tag{6.1}$$

The coordinates q_i by themselves therefore do not form a phase space. They describe configurations, not complete states. A genuine phase-space description must contain everything needed to say where the system goes next, and it must do so in a way that does not lose or create phase-space points.

6.2.1 Question & Answer

Question: Why are ordinary $F = ma$ equations not already of Hamiltonian form?

Answer: Because $F = ma$ is second order in time. A second-order law does not determine the future from the coordinates alone. It determines the future only after we also specify the first time derivatives. So the configuration variables q_i are not yet the full state. Hamiltonian mechanics begins by enlarging the description from configuration space to phase space.

Motivation. A student then asks the historical question that naturally arises here: why bother reformulating Newtonian mechanics at all? Susskind's answer has two layers. Lagrange's motivation was practical. One wants equations that do not depend on a specially chosen Cartesian coordinate system. Motion on a plane may be well described by x and y, but motion on a sphere is not. Hamilton's motivation, as Susskind presents it, is subtler: he seems to have seen a beautiful and extraordinarily elegant structure. Its deeper usefulness became unmistakable later, especially when quantum mechanics appeared. The lecture does not try to reconstruct Hamilton's private intuition; it asks us instead to follow the construction and then see why it matters.

6.3 From Second-Order Dynamics To First-Order Flow

Once we accept that the state must include both position and something velocity-like, the next step is straightforward. A system of second-order equations can be rewritten as twice as many first-order equations. The board work at this point is garbled in the transcript, so it is safest to present the standard form of the argument while keeping the lecture's point intact.

For a particle with coordinates x_i, define the momenta by

$$p_i = m\dot{x}_i. \tag{6.2}$$

Then Newton's equations become

$$\dot{x}_i = \frac{p_i}{m}, \tag{6.3}$$

$$\dot{p}_i = F_i. \tag{6.4}$$

We now have twice as many equations, but each one is only first order in time. The variables have doubled as well: the state is no longer the collection of positions x_i alone, but the collection of pairs (x_i, p_i). In general we write these as canonical pairs (q_i, p_i).

Susskind is careful to insist that this is not new physics. No new law of nature has been added. We have only changed the language in which the old law is expressed. But it is exactly the right language for the Hamiltonian point of view. A first-order system is the kind of system that tells us where to go next once we know where we are, and now "where we are" means a point in phase space.

This is the place in the lecture where the word *flow* becomes natural. If many identical systems start at many different points of phase space, then their collective evolution sweeps out a motion of the whole phase-space cloud. Hamiltonian mechanics is, from this point on, the study of that flow.

6.4 The Legendre Transformation

Only after that conceptual preparation does the lecture introduce the formal tool it needs. Susskind calls it a small piece of mathematics, easy but formal: the Legendre transformation.

We begin with a single pair of variables V and P. Secretly one may think of velocity and momentum, but for the moment they are just abstract variables. The crucial assumption is that they determine one another uniquely. In other words, P is a one-to-one function of V, and V is the inverse one-to-one function of P. The graph may not double back. If it did, one variable would cease to determine the other.

Now define a function $L(V)$ by

$$\frac{dL}{dV} = P. \tag{6.5}$$

Equivalently,

$$L(V) = \int_0^V P(V')\, dV'. \tag{6.6}$$

The lower limit 0 is just a convenient reference point; it fixes an additive constant and nothing more.

By symmetry we introduce a second function $H(P)$ satisfying

$$\frac{dH}{dP} = V, \qquad H(P) = \int_0^P V(P')\, dP'. \tag{6.7}$$

Geometrically, $L(V)$ is the area under the curve $P(V)$ up to the point V, while $H(P)$ is the complementary area defined by the same curve when P is treated as the independent variable. Together they fill a rectangle of area PV. Thus

$$H + L = PV, \qquad H = PV - L. \tag{6.8}$$

The reciprocity can be proved algebraically in one line. Start from $H = PV - L$ and vary P:

$$\begin{aligned} \delta H &= \delta(PV - L) && (6.9)\\ &= P\,\delta V + V\,\delta P - \frac{dL}{dV}\,\delta V && (6.10)\\ &= P\,\delta V + V\,\delta P - P\,\delta V && (6.11)\\ &= V\,\delta P. && (6.12) \end{aligned}$$

Therefore

$$\frac{dH}{dP} = V. \tag{6.13}$$

This is the Legendre transformation in the lecture's form. We begin with a function $L(V)$, differentiate it to define the conjugate variable P, and then build the transformed function $H(P) = PV - L$. The derivative relation reverses itself.

For many variables the pattern is the same. If $L = L(v_1, \ldots, v_n)$, then

$$\frac{\partial L}{\partial v_i} = p_i. \tag{6.14}$$

Define

$$H = \sum_i p_i v_i - L. \tag{6.15}$$

Varying this and cancelling the terms proportional to δv_i yields

$$\frac{\partial H}{\partial p_i} = v_i. \tag{6.16}$$

The lecture briefly remarks that thermodynamics is full of such Legendre pairs. But the real interest here is mechanics.

6.5 From The Abstract Transform To Mechanics

At this stage Susskind makes the decisive pivot. The abstract V–P story is now to be interpreted mechanically, with one important caveat: the coordinates q_i simply go along for the ride. The Legendre transformation is being carried out with respect to velocities, while the coordinates are treated as passive parameters.

The board evidence is documentary here, and it is worth preserving. In the lecture this stage is written as

$$\frac{\partial \mathcal{L}}{\partial V_i} = P_i, \qquad H(P) = \sum_i P_i V_i - \mathcal{L}. \tag{6.17}$$

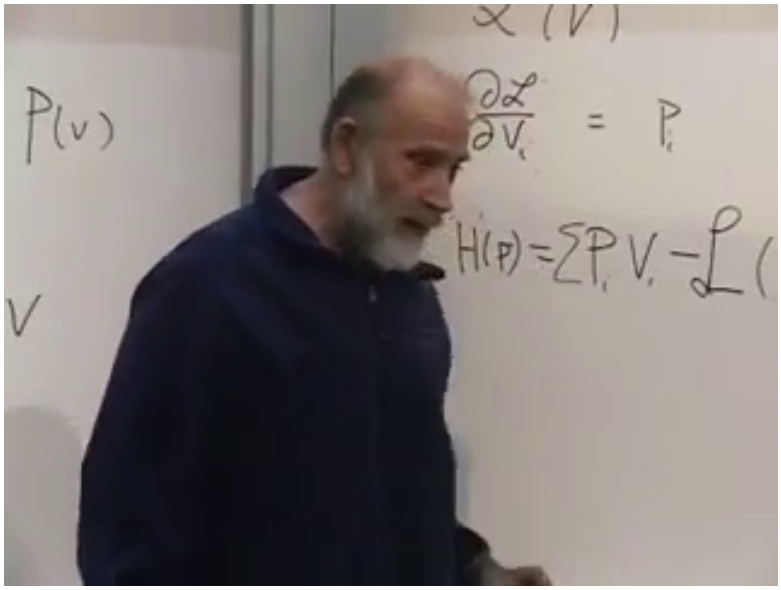


Figure 6.1: Hamiltonian as the Legendre transform of the Lagrangian. The board preserves the abstract notation $V_i, P_i, \mathcal{L}$ at the moment when the construction is carried into mechanics.

The frame does not certify more than this, so we keep exactly these visible equations and normalize the notation only in the surrounding prose.

Now return to mechanics. Let

$$v_i = \dot{q}_i. \tag{6.18}$$

Then the Lagrangian is a function of both coordinates and velocities:

$$L = L(q, \dot{q}, t) = L(q, v, t). \tag{6.19}$$

The canonical momenta are defined by

$$p_i = \frac{\partial L(q, \dot{q}, t)}{\partial \dot{q}_i} = \frac{\partial L(q, v, t)}{\partial v_i}. \tag{6.20}$$

This is exactly the mechanical realization of the Legendre-pair relation.

The Hamiltonian is then defined by

$$H(q, p, t) = \sum_i p_i \dot{q}_i - L(q, \dot{q}, t), \tag{6.21}$$

with the understanding that after the transform the velocities are to be expressed as functions of q, p, and t.

Here the one-to-one assumption returns in a concrete form. The relation between v_i and p_i must be invertible. If it is not, then H cannot be regarded as a well-defined function of q and p, and the Hamiltonian picture loses its grip.

6.6 Hamilton's Equations, Their Structure, And The Ordinary Particle

Now the formal heart of the lecture arrives. Starting from

$$H(q,p,t) = \sum_i p_i v_i - L(q,v,t), \tag{6.22}$$

with $v_i = \dot{q}_i$ understood as functions of q and p, we vary H. Since L depends on both q_i and v_i, while H is to be viewed as a function of q_i and p_i, the change is

$$\delta H = \sum_i (p_i\,\delta v_i + v_i\,\delta p_i) - \sum_i \frac{\partial L}{\partial q_i}\,\delta q_i - \sum_i \frac{\partial L}{\partial v_i}\,\delta v_i \tag{6.23}$$

$$= \sum_i v_i\,\delta p_i - \sum_i \frac{\partial L}{\partial q_i}\,\delta q_i. \tag{6.24}$$

The entire mechanism is in the cancellation: because

$$p_i = \frac{\partial L}{\partial v_i}, \tag{6.25}$$

the δv_i terms disappear.

From the remaining variation we read off

$$\frac{\partial H}{\partial p_i} = v_i = \dot{q}_i, \qquad \frac{\partial H}{\partial q_i} = -\frac{\partial L}{\partial q_i}. \tag{6.26}$$

Now use Lagrange's equations,

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{q}_i} = \frac{\partial L}{\partial q_i}. \tag{6.27}$$

Since $\partial L/\partial \dot{q}_i = p_i$, this becomes

$$\dot{p}_i = \frac{\partial L}{\partial q_i}. \tag{6.28}$$

Therefore

$$\dot{p}_i = -\frac{\partial H}{\partial q_i}. \tag{6.29}$$

So Hamilton's equations are

$$\dot{q}_i = \frac{\partial H}{\partial p_i}, \tag{6.30}$$

$$\dot{p}_i = -\frac{\partial H}{\partial q_i}. \tag{6.31}$$

This is where the lecture slows down and admires the structure. Coordinates and momenta enter almost symmetrically. The whole asymmetry is concentrated into one minus sign. Even more strikingly, the derivatives are crossed: the q-velocity comes from a p-derivative, and the p-velocity comes from a q-derivative. That crossing is not an ornament. It is the source of the geometric properties that come later.

A boundary question. Before checking the formalism on an ordinary particle, the lecture pauses for a caution. Any second-order system can certainly be rewritten as a first-order system by introducing new variables; that part is trivial. But the elegant Hamiltonian structure is more than that. It requires a genuine Legendre transform, and therefore a one-to-one relation between the velocity variables and the momentum variables. If that relation cannot be inverted, the motion ceases to be well defined in the Hamiltonian sense. The lecture also stresses that friction lies outside the reduced Hamiltonian description: friction appears only after many microscopic degrees of freedom have been ignored.

Now let us check the equations on familiar ground. Take the ordinary one-dimensional particle with

$$L = \frac{1}{2}m\dot{x}^2 - U(x). \tag{6.32}$$

Then

$$p = m\dot{x}, \qquad H = p\dot{x} - L = \frac{p^2}{2m} + U(x). \tag{6.33}$$

The first Hamilton equation gives

$$\frac{\partial H}{\partial p} = \frac{p}{m} = \dot{x}, \tag{6.34}$$

which is just the relation between momentum and velocity.

The second gives

$$\dot{p} = -\frac{\partial H}{\partial x} = -\frac{dU}{dx}. \tag{6.35}$$

Since $p = m\dot{x}$, we recover

$$m\ddot{x} = -\frac{dU}{dx}. \tag{6.36}$$

So the formalism does work. Typically, as the lecture emphasizes, the first Hamilton equation is the kinematic relation between momentum and velocity, while the second is the one that looks like Newton's equation.

At this point the reader is meant to feel the subject shift. Mechanics has become a system of first-order equations for canonical pairs. Given a single function $H(q, p)$, differentiation tells us the phase-space velocity of the system.

6.7 Energy, Contours, And General Conservation Laws

Having checked the equations on the ordinary particle, the lecture turns to the first major payoff: energy conservation. If the Hamiltonian has no explicit time dependence, then

$$\frac{dH}{dt} = \sum_i \left(\frac{\partial H}{\partial p_i}\dot{p}_i + \frac{\partial H}{\partial q_i}\dot{q}_i \right). \tag{6.37}$$

Now substitute Hamilton's equations:

$$\frac{dH}{dt} = \sum_i \left(\frac{\partial H}{\partial p_i}\left(-\frac{\partial H}{\partial q_i}\right) + \frac{\partial H}{\partial q_i}\left(\frac{\partial H}{\partial p_i}\right) \right) \tag{6.38}$$

$$= 0. \tag{6.39}$$

Hence

$$\frac{dH}{dt} = 0. \tag{6.40}$$

Susskind pauses over how simple this is. Energy conservation is not here an added theorem sitting on top of the dynamics. It is a direct cancellation inside the Hamiltonian form of the equations. He also notes the logical direction carefully: Hamiltonian equations with a time-independent Hamiltonian imply conservation of energy, but not every energy-conserving system need be Hamiltonian.

Now the geometric meaning becomes vivid. Think of phase space as filled with contours of constant energy. As the phase point moves in time, it remains on the contour on which it started. The phase-space dust flows along those constant-energy curves.

For the harmonic oscillator, suppressing inessential constants,

$$H \propto p^2 + x^2, \tag{6.41}$$

so the constant-energy contours are simply

$$p^2 + x^2 = \text{constant}, \tag{6.42}$$

namely circles around the origin. The origin itself is a degenerate circle: the oscillator sits there motionless.

The contours need not all be closed. The lecture stresses this with simple examples. Small oscillations in a periodic well give closed curves. A particle with enough energy to run over a washboard potential gives open curves. The same phase space may therefore contain both bounded and unbounded motions.

6.7.1 Question & Answer

Question: If the phase-space fluid is incompressible, how can the flow lines bunch up or spread out?

Answer: The lecture answers this with the image of a fluid going through a narrow channel. The fluid need not compress. It may simply move faster where the channel narrows and slower where it widens. The same is true here. When the constant-energy contours crowd together, the derivatives of H are large, and therefore the phase-space velocity

$$(\dot{q}_i, \dot{p}_i) \tag{6.43}$$

is large. Where the contours spread apart, the motion slows down. The bunching of lines is compensated by speed, not by a change in density.

This answer also meshes perfectly with Hamilton's equations. Close contours mean that H changes rapidly with p and q, so the corresponding derivatives are large, which is exactly what makes the phase-space point move rapidly.

The lecture now widens from energy to general conservation laws. Let $A(q, p)$ be any quantity on phase space. Then

$$\frac{dA}{dt} = \sum_i \left(\frac{\partial A}{\partial p_i} \dot{p}_i + \frac{\partial A}{\partial q_i} \dot{q}_i \right). \tag{6.44}$$

Substituting Hamilton's equations gives

$$\frac{dA}{dt} = \sum_i \left(\frac{\partial A}{\partial q_i} \frac{\partial H}{\partial p_i} - \frac{\partial A}{\partial p_i} \frac{\partial H}{\partial q_i} \right). \tag{6.45}$$

This suggests introducing a new piece of structure. In the convention we adopt here, following the lecture as reconstructed from the transcript,

$$\{A, B\} = \sum_i \left(\frac{\partial A}{\partial q_i} \frac{\partial B}{\partial p_i} - \frac{\partial A}{\partial p_i} \frac{\partial B}{\partial q_i} \right). \tag{6.46}$$

There are sign conventions in the literature, and the late discussion in the lecture is conversational and somewhat tangled, but one must make a convention, and this is the one used here.

With that definition,

$$\frac{dA}{dt} = \{A, H\}. \tag{6.47}$$

This is already a new formulation of mechanics. The Hamiltonian is no longer just the energy function written in new variables. It is the generator of time evolution through the Poisson bracket.

Hamilton's equations themselves are recovered as special cases. If $A = q_i$, then

$$\{q_i, H\} = \dot{q}_i. \tag{6.48}$$

If $A = p_i$, then

$$\{p_i, H\} = \dot{p}_i. \tag{6.49}$$

So the Poisson-bracket law is indeed a generalization of Hamilton's equations.

A quantity is conserved precisely when its time derivative vanishes. Therefore

$$\{A, H\} = 0 \tag{6.50}$$

is the necessary and sufficient condition for A to be conserved. For instance, the angular momentum of a particle moving in a plane,

$$L_z = xp_y - yp_x, \tag{6.51}$$

is conserved exactly when its Poisson bracket with the Hamiltonian vanishes.

The lecture then adds one last refinement. If the Hamiltonian depends explicitly on time, then the chain rule gives

$$\frac{dH}{dt} = \sum_i \left(\frac{\partial H}{\partial p_i} \dot{p}_i + \frac{\partial H}{\partial q_i} \dot{q}_i \right) + \frac{\partial H}{\partial t}. \tag{6.52}$$

The first two terms still cancel, so

$$\frac{dH}{dt} = \frac{\partial H}{\partial t}. \tag{6.53}$$

This is the clean result the lecture wants. If an external source moves and thereby changes the potential even while the system's own coordinates are held fixed, then the Hamiltonian has explicit time dependence, and energy need not be conserved within the subsystem we chose to isolate.

6.8 Summary

Hamiltonian mechanics is introduced in this lecture not as a bag of formulas, but as a disciplined way of thinking about reversible evolution. The state must be rich enough to determine both future and past, and that requirement forces us from configuration space to phase space. The Legendre transformation then changes the dynamical variables from velocities to canonical momenta, so that the motion is encoded in a single function $H(q, p)$.

Once that change has been made, mechanics takes the remarkably compact form

$$\dot{q}_i = \frac{\partial H}{\partial p_i}, \qquad \dot{p}_i = -\frac{\partial H}{\partial q_i}. \tag{6.54}$$

From there the main payoffs unfold exactly in the order of the lecture. The ordinary particle is recovered at once. Energy conservation becomes a direct cancellation. Motion is reinterpreted geometrically as flow along constant-energy contours. And finally the whole picture widens into the Poisson-bracket language, where the Hamiltonian generates the time dependence of every observable quantity.

By the end of the lecture, Hamiltonian mechanics has been shown in three complementary ways at once: as a reformulation of Newtonian dynamics, as a geometry of phase-space flow, and as a general algebraic language for conservation laws.